

Multimodality Contd., Ethics in NLP

CSE 5525: Foundations of Speech and Natural Language
Processing

<https://shocheen.github.io/courses/cse-5525-fall-2025>



THE OHIO STATE UNIVERSITY

Recap: Why do we want to build visual multimodal models?

- . Image understanding
- . Image Generation (won't cover in this course)
- . Improve text understanding / generation?

Vision Transformer (ViT)



[An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale](https://arxiv.org/abs/2010.11929)

[Tutorial 11: Vision Transformers](#)

Figure: <https://github.com/lucidrains/vit-pytorch/blob/main/images/vit.gif>

Vision Transformer (ViT)

We learned that pretraining helps!



[An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale](#)

[Tutorial 11: Vision Transformers](#)

Figure: <https://github.com/lucidrains/vit-pytorch/blob/main/images/vit.gif>

CLIP

[Radford et al., 2021]; [Conference presentation](#)

– Contrastive pretraining pseudocode

```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

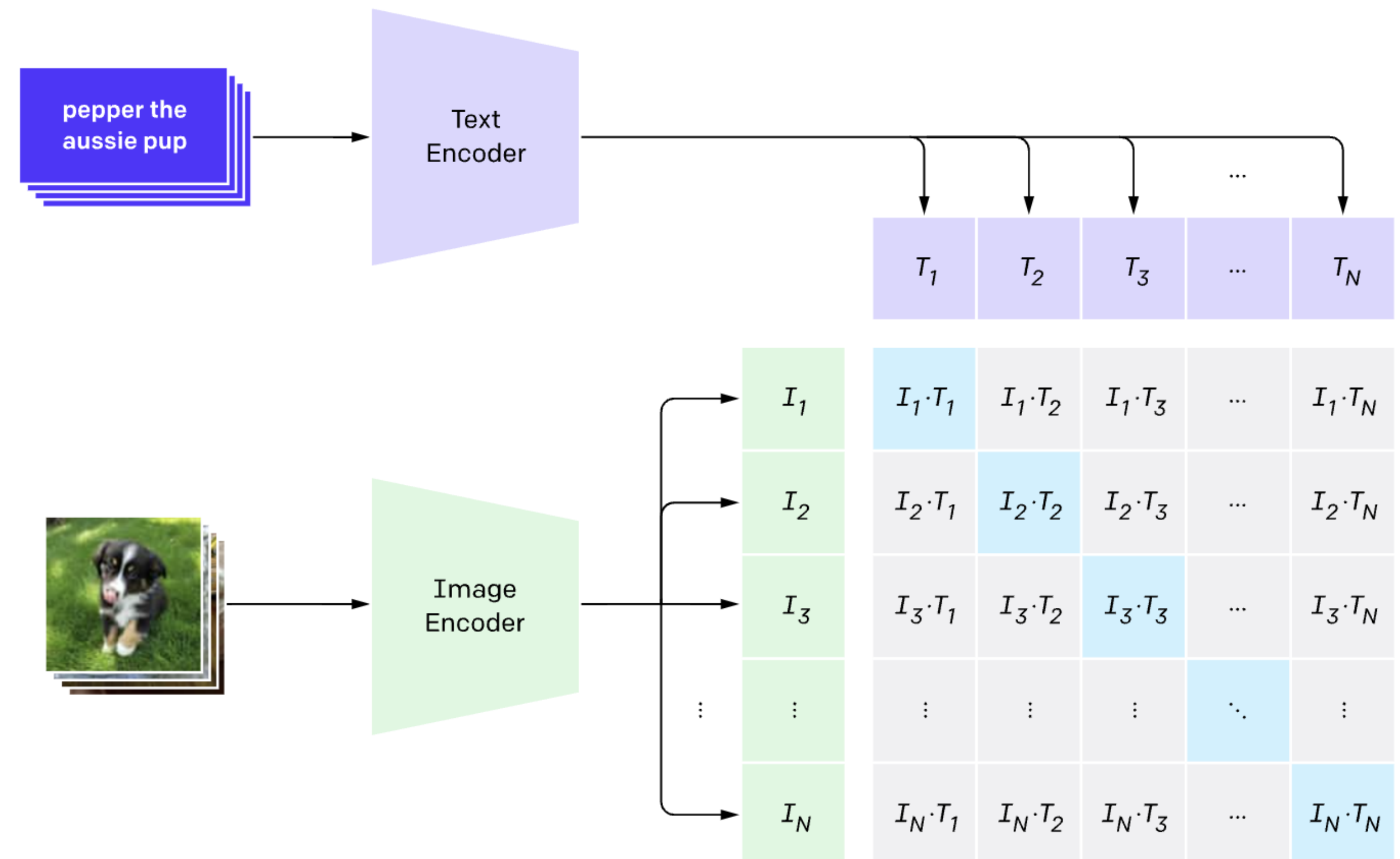
# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

Figure 3. Numpy-like pseudocode for the core of an implementation of CLIP.

1. Contrastive pre-training



CLIP

[[Radford et al., 2021](#)]; [Conference presentation](#)

Original repository, zero-shot prediction:

<https://github.com/openai/CLIP#zero-shot-prediction>

In 🤗 ecosystem:

https://huggingface.co/docs/transformers/model_doc/clip

Independently trained and larger CLIP:

https://github.com/mlfoundations/open_clip

Goals of Today's Lecture

Goal: Learn how some LLMs that take more than just text

- ↓ Motivation for V&L models
- ↓ Vision Transformer
- ↓ **Classification with Image+Text Input**
- ↓ Generation with Image+Text Input
- ↓ Video Processing
- ↓ Speech Processing

Multimodal Classification / Visual Question Answering



Q: What endangered animal is featured on the truck?

- A: **A bald eagle.**
A: A sparrow.
A: A hummingbird.
A: A raven.



Q: Where will the driver go if turning right?

- A: **Onto 24 3/4 Rd.**
A: Onto 25 3/4 Rd.
A: Onto 23 3/4 Rd.
A: Onto Main Street.



Q: When was the picture taken?

- A: **During a wedding.**
A: During a bar mitzvah.
A: During a funeral.
A: During a Sunday church service.



Q: Who is under the umbrella?

- A: **Two women.**
A: A child.
A: An old man.
A: A husband and a wife.



Q: Why was the hand of the woman over the left shoulder of the man?

- A: **They were together and engaging in affection.**
A: The woman was trying to get the man's attention.
A: The woman was trying to scare the man.
A: The woman was holding on to the man for balance.



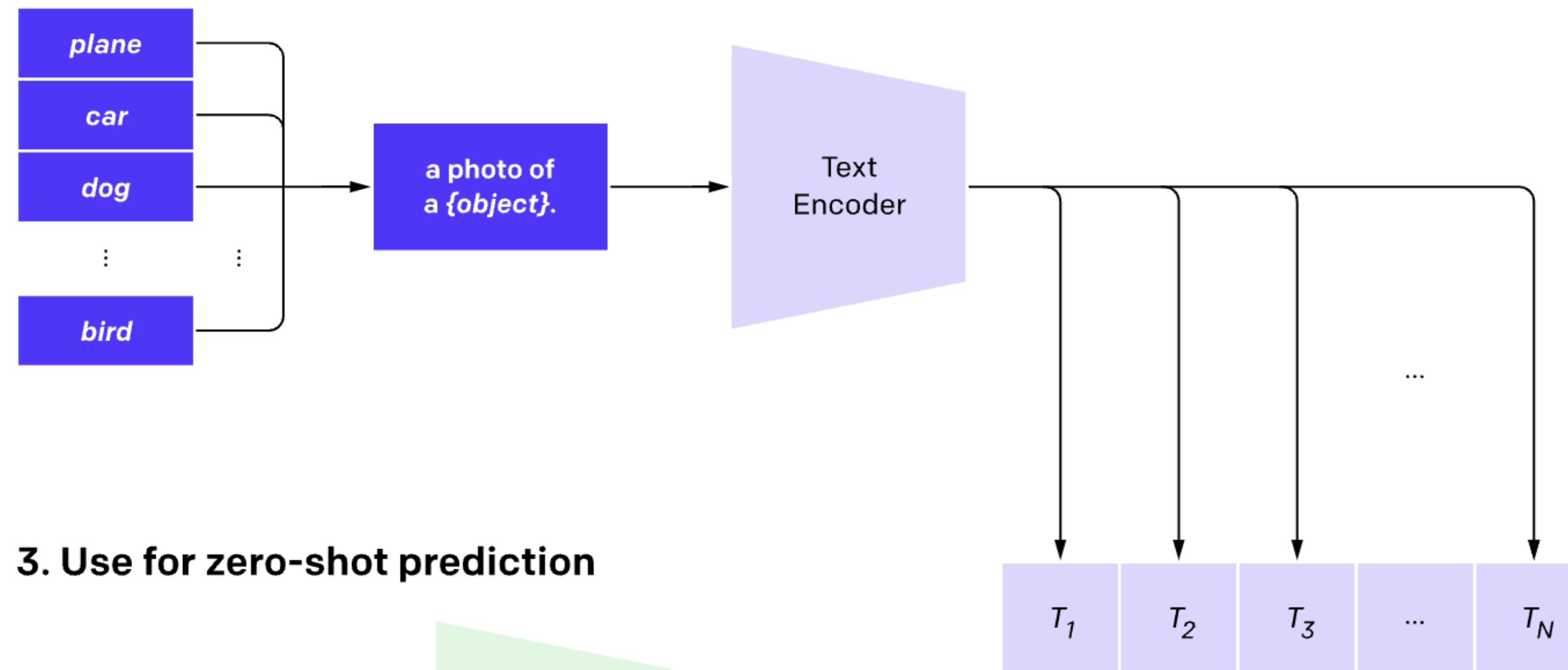
Q: How many magnets are on the bottom of the fridge?

- A: **5.**
A: 2.
A: 3.
A: 4.

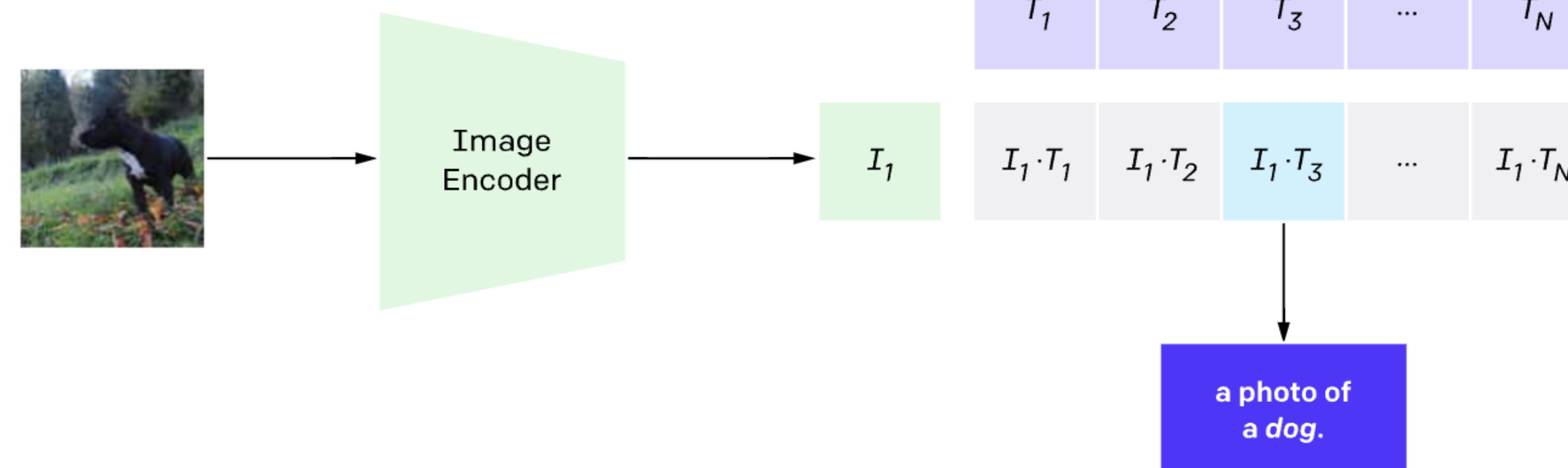
CLIP [\[Radford et al., 2021\]](#); [Conference presentation](#)

– *Image classification*

2. Create dataset classifier from label text



3. Use for zero-shot prediction



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- ↳ **Generation with Image+Text Input**
- ↳ Video Processing
- ↳ Speech Processing

Not constrained to classification

User What is funny about this image? Describe it panel by panel.



Source: hmmm (Reddit)

GPT-4o: Not constrained to classification

What is funny about this image? Describe

GPT-4

The image shows a package for a "Lightning Cable" adapter with three panels.

Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port.

Panel 2: The package for the "Lightning Cable" adapter with a picture of a VGA connector on it.

Panel 3: A close-up of the VGA connector with a small Lightning connector (used for charging iPhones and other Apple devices) at the end.

The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.



Source: [hmmm \(Reddit\)](#)

Components of a simple and standard design of combining a language model with a vision encoder

Image encoder:

- ↳ *Image preprocessing*: Turn an image into a sequence of patches
- ↳ A *pretrained* Vision Transformer image encoder that first maps each of image patches into input embeddings, then transforms them using many self-attention and FF/MLP layers

Cross modal connector

- ↳ A connector that projects the vision embeddings (from e.g. final layer) to the language model's input dimension with an FFNN/MLP
- ↳ Initially randomly initialized

A *pretrained* decoder-only Transformer LLM

- ↳ Prepend projected vision embeddings to the token embeddings

LLaVA: Visual Instruction Tuning <https://llava-vl.github.io/>

Strong pretrained vision and language models

- Vision encoder: CLIP-ViT-L/14
- Language model: LLaMA-2, etc.

Cross modal connector

- Linear projection

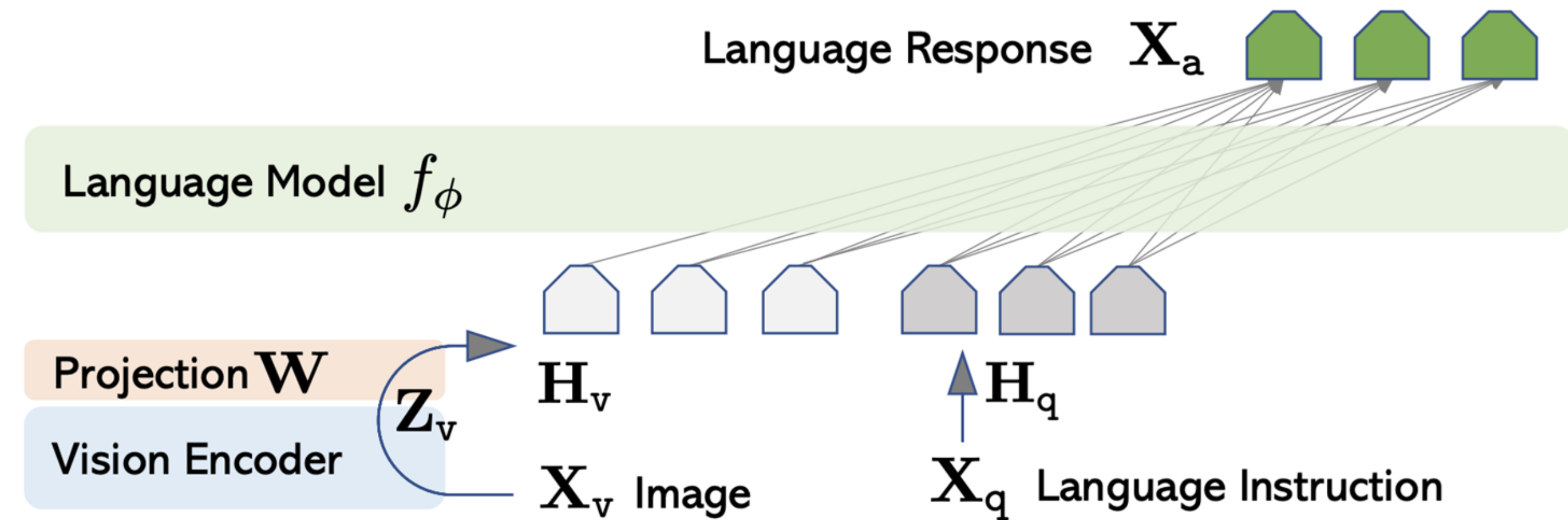


Figure 1: LLaVA network architecture.

Tuning the model for following multimodal instructions

- Use image captions from available datasets
- Prompt text-only GPT-4 to generate (instruction, output) pairs
- 158K instructions

First tune only the projection, then tune the projection and LM

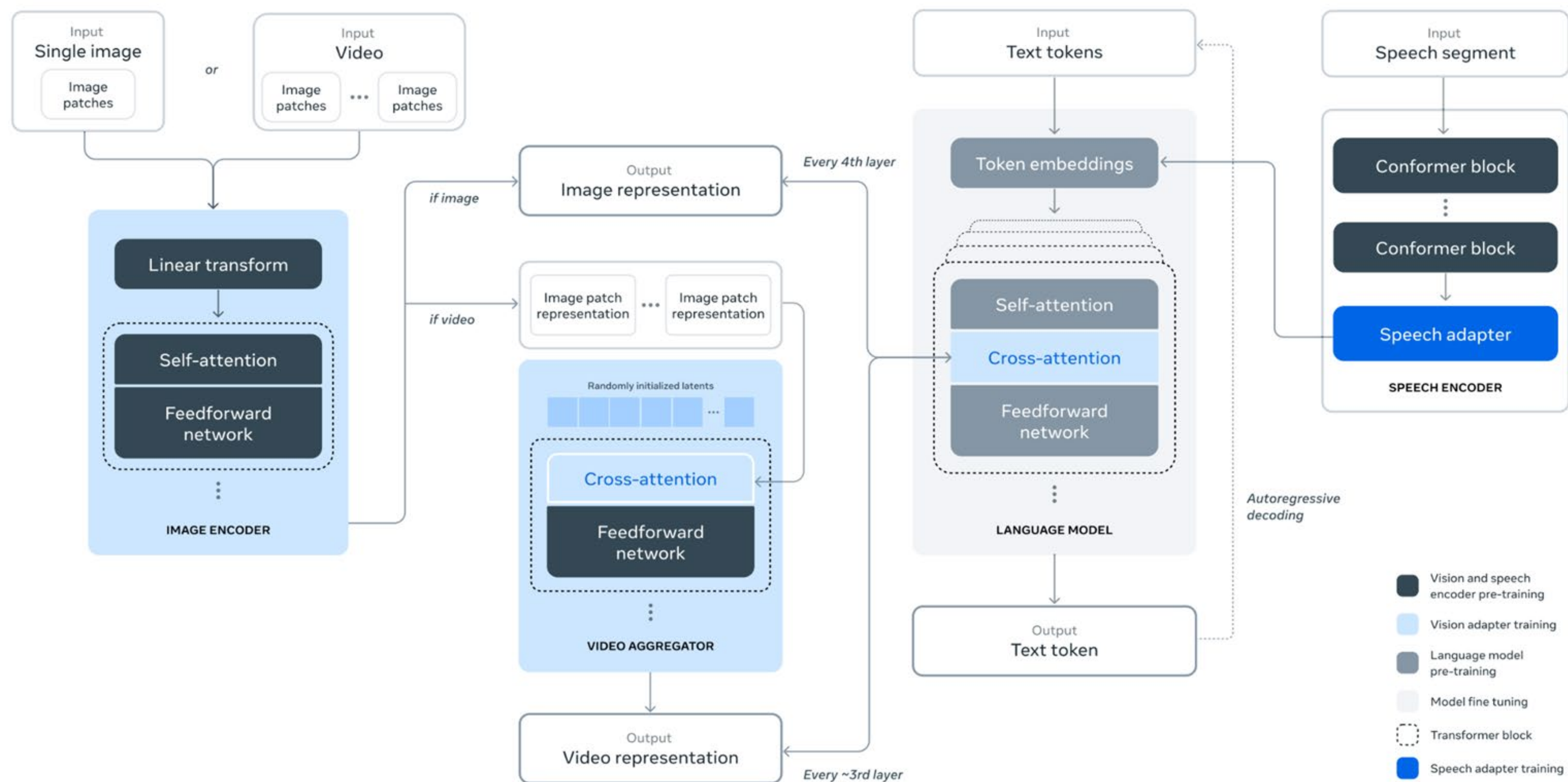
Category	Model	VLM		LLM Backbone		Vision Encoder	
		Open Weights	Open Data + Code	Open Weights	Open Data + Code	Open Weights	Open Data + Code
Molmo	Molmo-72B	Open	Open	Open	Closed	Open	Closed
	Molmo-7B-D	Open	Open	Open	Closed	Open	Closed
	Molmo-7B-O	Open	Open	Open	Open	Open	Closed
	MolmoE-1B	Open	Open	Open	Open	Open	Closed
API Models	GPT-4o	Closed	Closed	Closed	Closed	Closed	Closed
	GPT-4V	Closed	Closed	Closed	Closed	Closed	Closed
	Gemini 1.5 Pro	Closed	Closed	Closed	Closed	Closed	Closed
	Gemini 1.5 Flash	Closed	Closed	Closed	Closed	Closed	Closed
	Claude 3.5 Sonnet	Closed	Closed	Closed	Closed	Closed	Closed
	Claude 3 Opus	Closed	Closed	Closed	Closed	Closed	Closed
	Claude 3 Haiku	Closed	Closed	Closed	Closed	Closed	Closed
Open Weights	Qwen VL2 72B	Open	Closed	Open	Closed	Open	Closed
	Qwen VL2 7B	Open	Closed	Open	Closed	Open	Closed
	Intern VL2 LLAMA 76B	Open	Closed	Open	Closed	Open	Closed
	Intern VL2 8B	Open	Closed	Open	Closed	Open	Closed
	Pixtral 12B	Open	Closed	Open	Closed	Open	Closed
	Phi3.5-Vision 4B	Open	Closed	Open	Closed	Open	Closed
	PaliGemma 3B	Open	Closed	Open	Closed	Open	Closed
Open Weights & Data	LLAVA OneVision 72B	Open	Distilled	Open	Closed	Open	Closed
	LLAVA OneVision 7B	Open	Distilled	Open	Closed	Open	Closed
	Cambrian-1 34B	Open	Distilled	Open	Closed	Open	Closed
	Cambrian-1 8B	Open	Distilled	Open	Closed	Open	Closed
	xGen - MM - Interleave 4B	Open	Distilled	Open	Closed	Open	Closed
	LLAVA-1.5 13B	Open	Open	Open	Closed	Open	Closed
	LLAVA-1.5 7B	Open	Open	Open	Closed	Open	Closed

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- ↳ **Video Processing**
- ↳ Speech Processing

Llama 3.2



Llama 3.2 (cont.)

Image encoder:

- Vision Transformer pretrained from scratch
- 224 x 224 resolution; 14 x 14 patches
- The size of patch embeddings = 7680
- Features from the 4th, 8th, 16th, 24th and 31st layers are also provided in addition to the final layer features

Cross modal connector:

- Cross-attention
- Introduce substantial numbers of additional trainable parameters into the model:
for Llama 3 405B, the cross-attention layers have $\approx 100\text{B}$ parameters

Language model: Llama 3.1

Llama 3.2 – Video processing

Llama 3.2 takes as input up to 64 uniformly sampled frames from a full video

Each frame is processed by the image encoder

Temporal structure in videos through two components:

1. Encoded video frames are aggregated by a temporal aggregator which merges 32 consecutive frames into one [[Jaegle et al., 2021](#)]
2. Extra video cross attention layers are added before every 4th image cross attention layer

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- ↳ **Speech Processing**

We haven't covered speech in class...

Analog signal

Goal: Raw wavefile $\Rightarrow$ Sequences of log mel spectrum vectors

Raw wavefile contains info about changes in air pressure caused the specific way that air passes through the glottis [the middle region inside your voice box that contains your vocal cords] & out the oral or nasal cavities

The graph measures the amount of **compression** or **rarefaction** (uncompression) of the air molecules

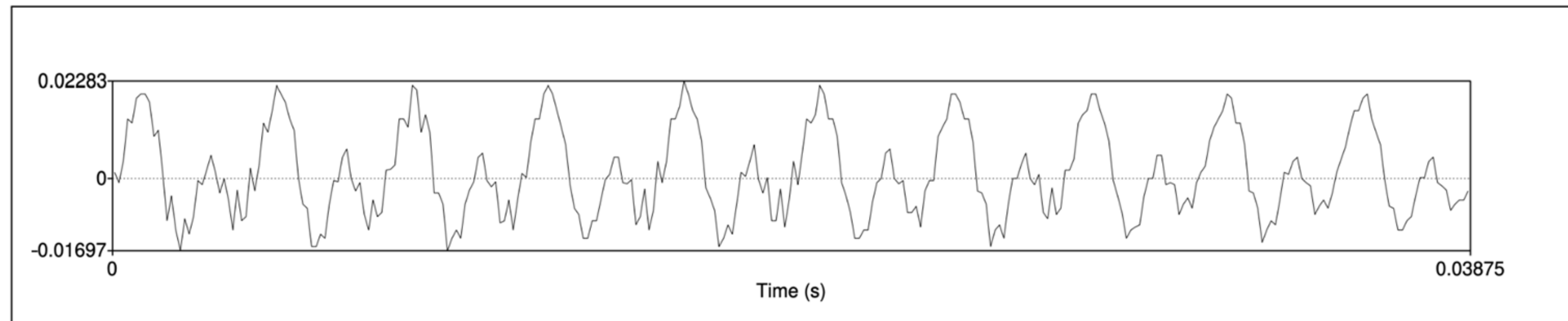


Figure 16.2 A waveform of an instance of the vowel [iy] (the last vowel in the word “baby”). The y-axis shows the level of air pressure above and below normal atmospheric pressure. The x-axis shows time. Notice that the wave repeats regularly.

Sampling and Quantization

Next steps: Transform a waveform, a 2D plot of air pressure changes (y-axis) over time (x-axis) into a sequence of 80-dimensional log Mel spectrum vectors

Sampling:

- Turn a waveform into a sequence of amplitude values [loudness] sampled at regular intervals (e.g., 16 kHz)
- Sampling rate: Number of samples/sec (e.g., 16 kHz for high-quality audio)
- Creates a 1D array of sampled amplitudes

Quantization:

- Digital systems work with discrete values rather than continuous ones
- Represents amplitudes as integers (e.g., 8-bit or 16-bit)
- Reduces continuous signal values into discrete levels

Windowing

Speech analyzed in small stationary windows

- Assumption: within small time windows, the properties of a speech signal (such as its frequency content) remain relatively constant

Key parameters:

- Window size (e.g., 25 ms): The duration of the analyzed segment
- Frame stride (e.g., 10 ms): The interval at which consecutive windows are started $\Rightarrow$ overlapping analysis allowed

Window types:

- Rectangular: Abrupt cutoff at edges
- Hamming: Smooth tapering at edges

Windowing results in a 2D array where each row corresponds to the samples in a window

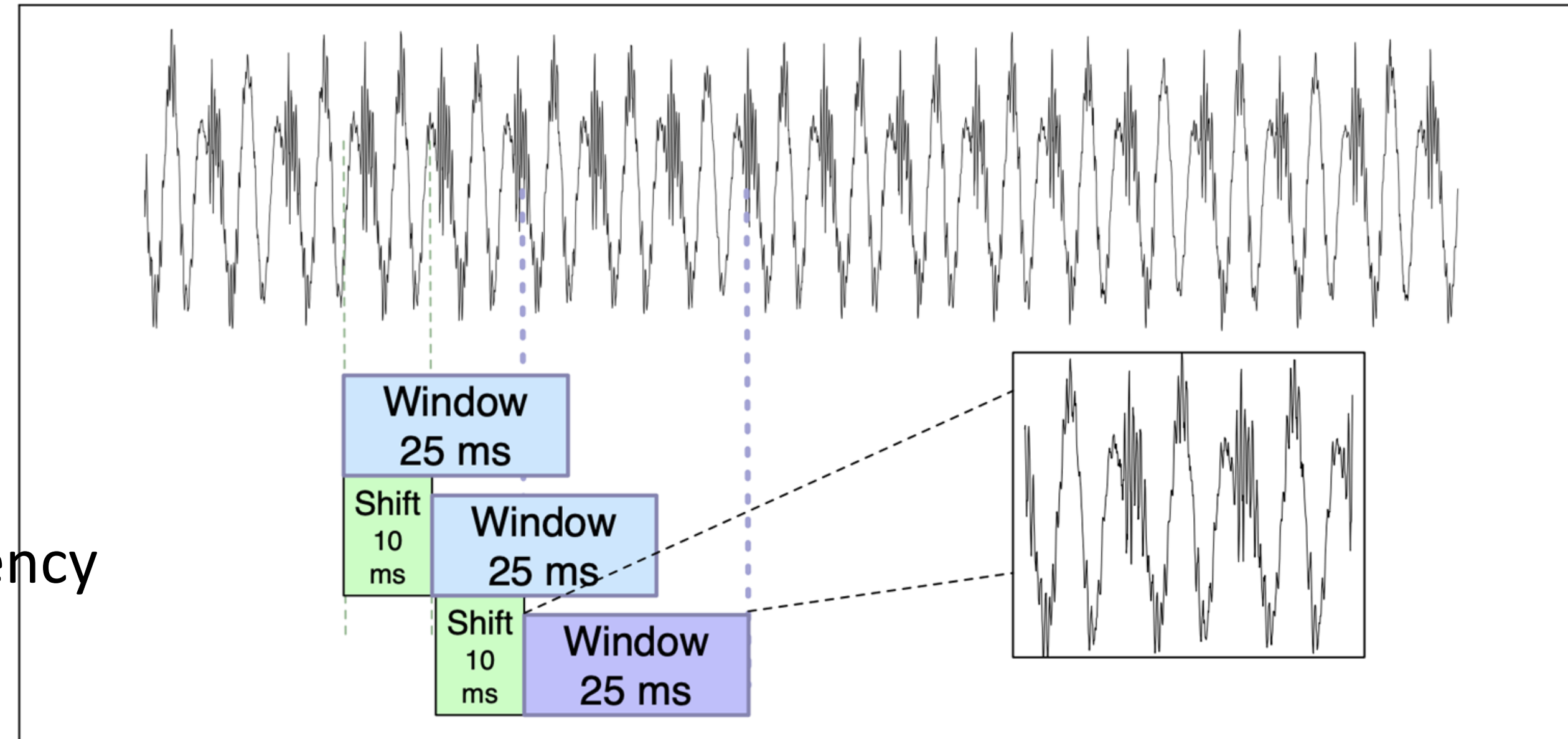


Figure 16.4 Windowing, showing a 25 ms rectangular window with a 10ms stride.

[[Jurafsky & Martin Section 16.2](#)]

Discrete Fourier Transform (DFT)

Next: Analyze the signal in the frequency domain rather than the time domain

A signal contains energy distributed across various frequencies

Spectral information: The breakdown of how much energy (or power) is present at each frequency band

Fast Fourier Transform (FFT): Efficient computation of DFT for signal analysis

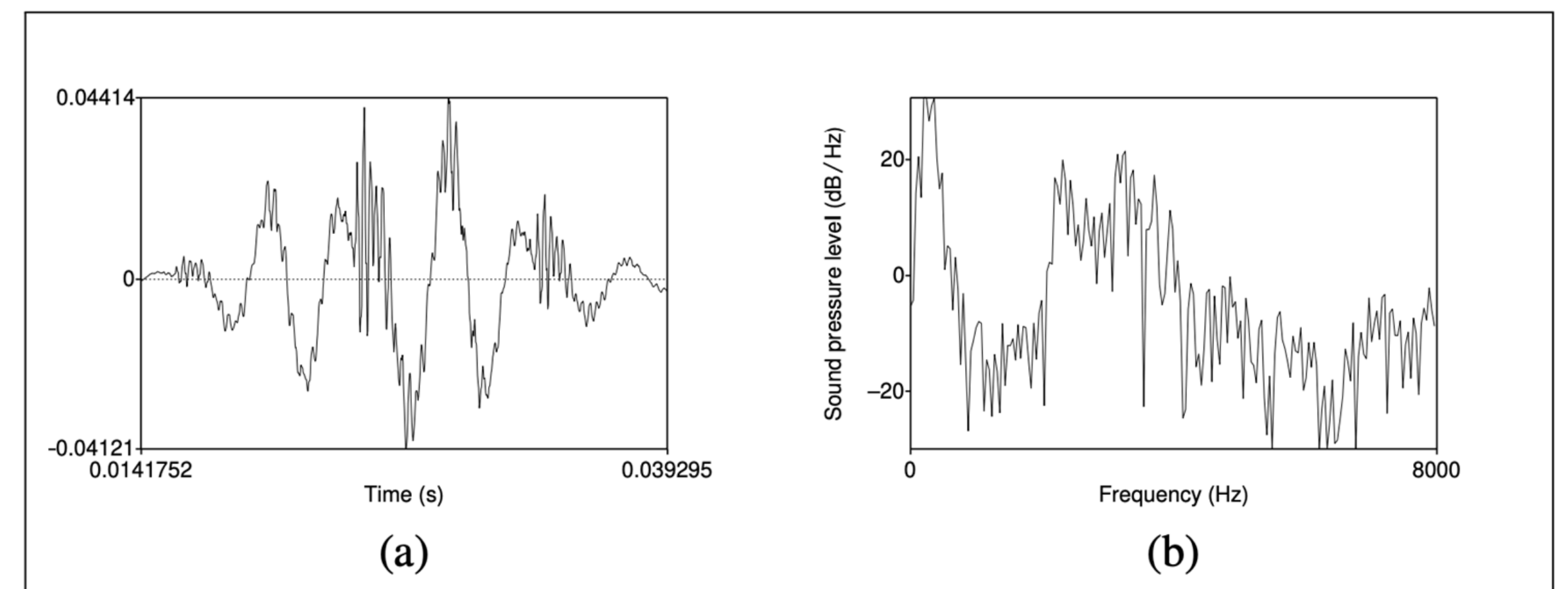


Figure 16.6 (a) A 25 ms Hamming-windowed portion of a signal from the vowel [iy] and (b) its spectrum computed by a DFT.

Mel Filter Bank

The results of the FFT tell us the energy at each frequency band

Human hearing is not equally sensitive at all frequency bands; it is less sensitive at higher frequencies

- This bias toward low frequencies helps human recognition, since information in low frequencies is crucial for distinguishing vowels or nasals, while information in high frequencies is less crucial for successful recognition

Mel is a unit of pitch [the degree of highness or lowness of a tone] $\Rightarrow$ Convert frequency to Mel scale

$$mel(f) = 1127 \ln\left(1 + \frac{f}{700}\right)$$

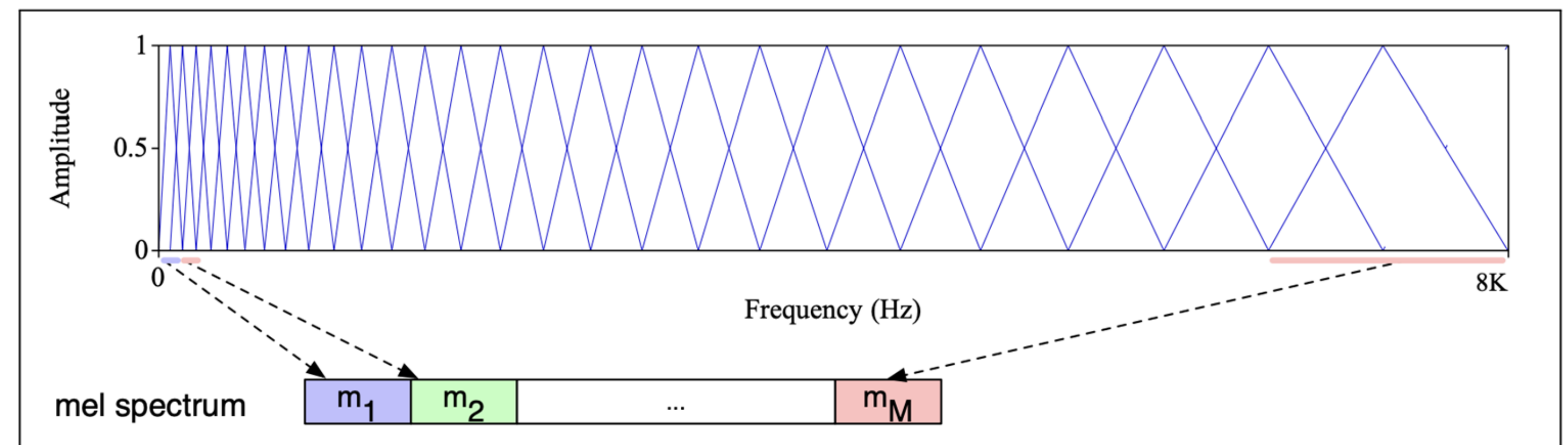


Figure 16.7 The mel filter bank (Davis and Mermelstein, 1980). Each triangular filter, spaced logarithmically along the mel scale, collects energy from a given frequency range.

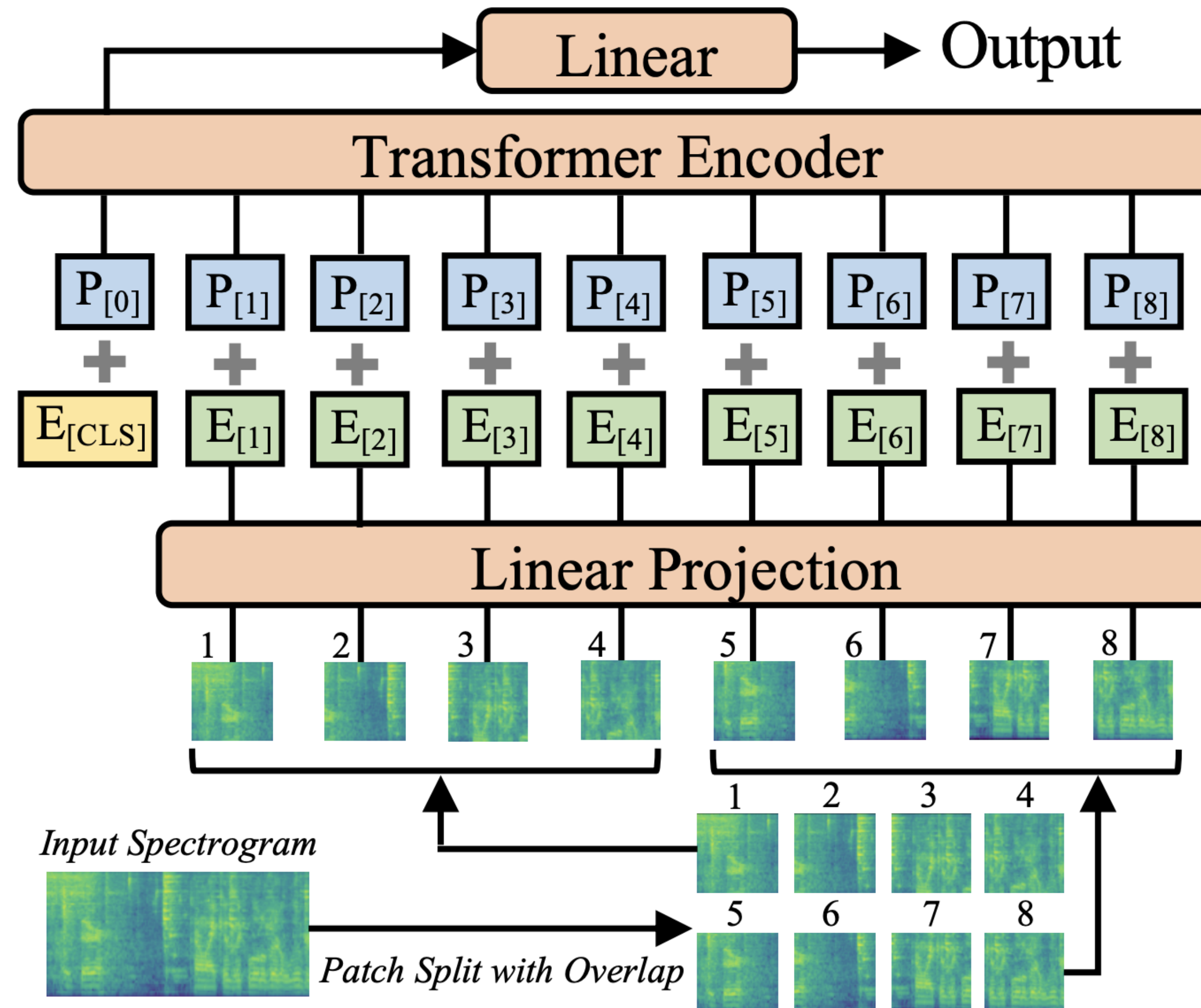
Log

The human response to signal level is logarithmic: Humans are less sensitive to slight differences in amplitude at high amplitudes than at low amplitudes

→ Take the log of each of the mel spectrum values!

Using a log also makes the feature estimates less sensitive to variations in input such as variations due to the speaker's mouth moving closer or further from the microphone

Audio Spectrogram Transformer [[Gong et al., 2021](#)]



Qwen2-Audio [[Chu et al., 2024](#)]

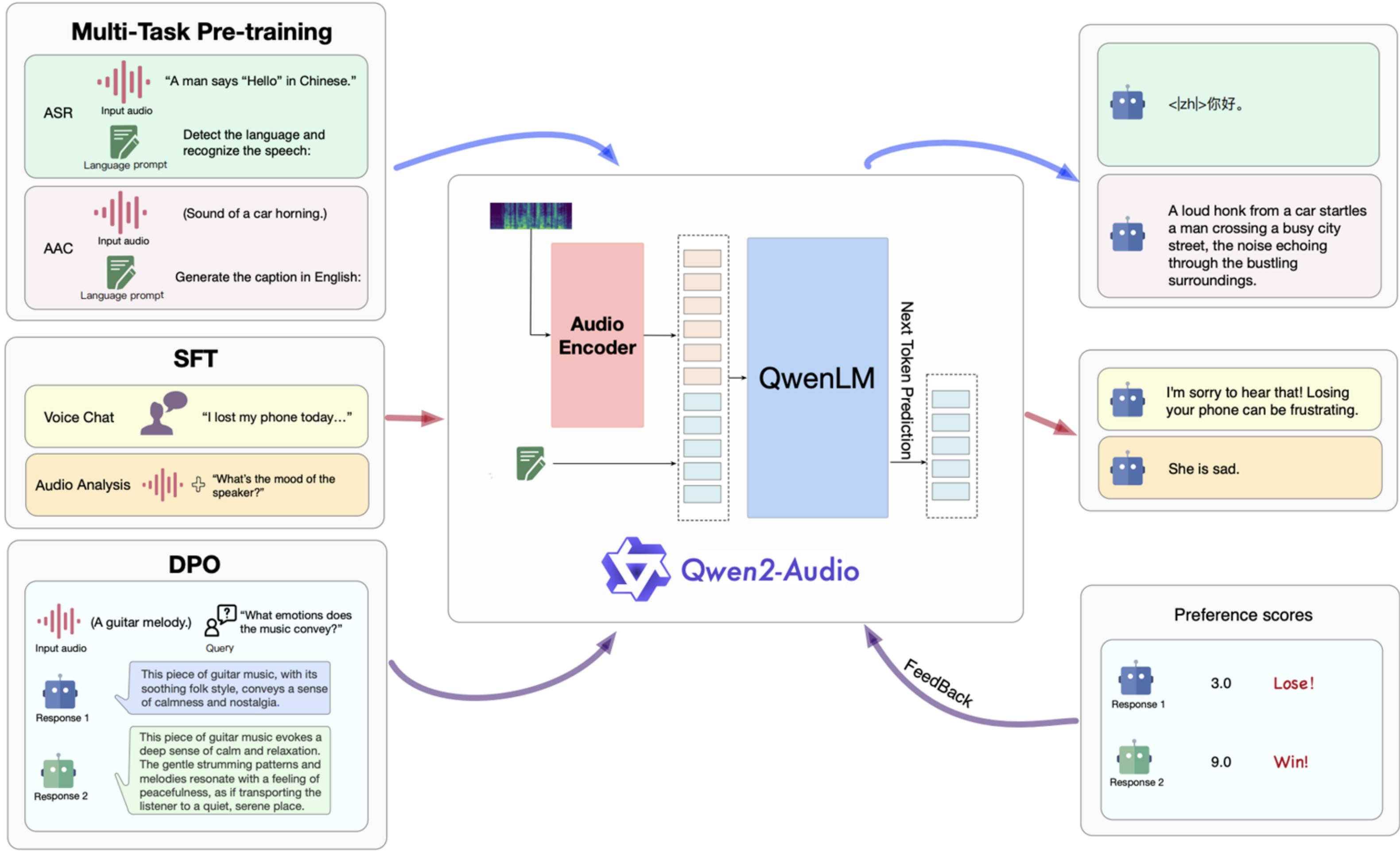


Figure 2: The overview of three-stage training process of Qwen2-Audio.

AI Ethics

What is Ethics?

Ethics is a study of what are **good and bad** ends to pursue in life and what it is **right and wrong** to do in the conduct of life.

It is therefore, above all, a practical discipline.

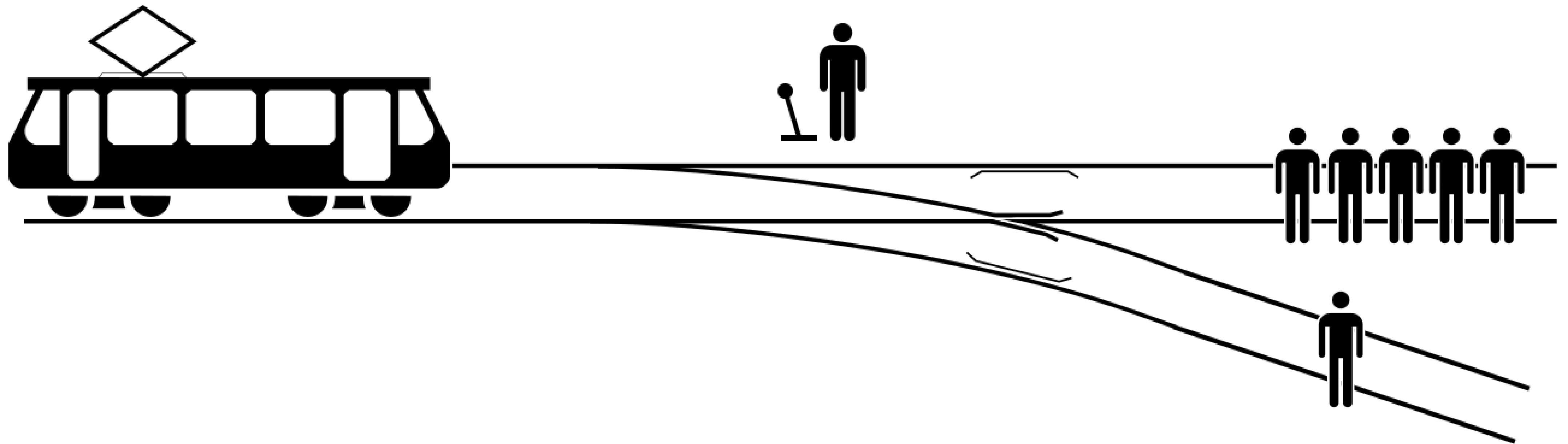
Its primary aim is to determine how one ought to live and what actions one ought to do in the conduct of one's life.”

What is Ethics?

It's the **good** things

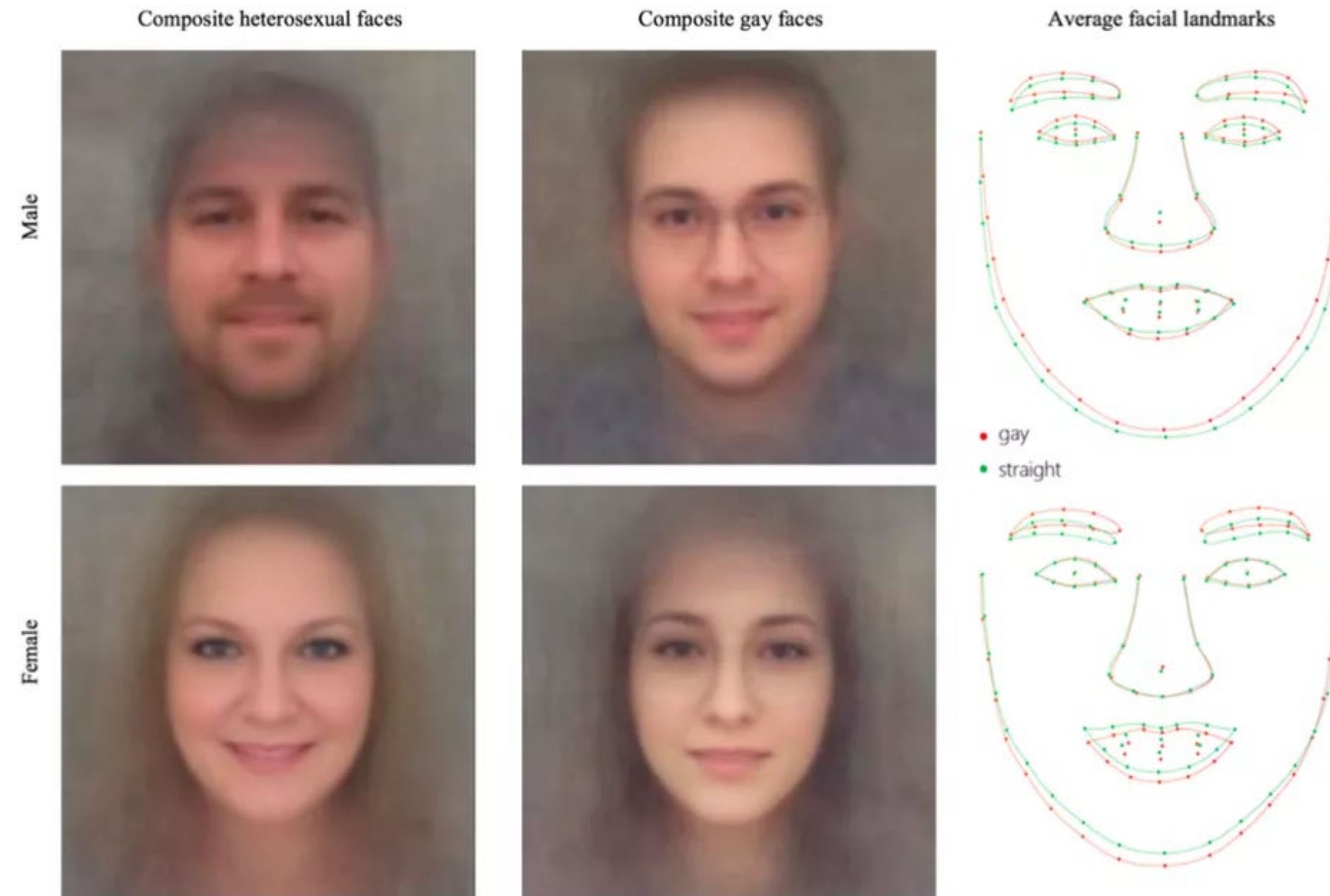
It's the **right** things

The Trolley Dilemma



What are important ethical questions to ask in development and deployment of AI systems?

A study: the “AI Gaydar”, 2017



A recent study: the “AI Gaydar”

- Research question
 - Identification of sexual orientation from facial features
- Data collection
 - Photos downloaded from a popular American dating website
 - 35,326 pictures of 14,776 people, all white, with gay and straight, male and female, all represented evenly
- Method
 - A deep learning model was used to extract facial features + grooming features; then a logistic regression classifier was applied for classification
- Accuracy
 - 81% for men, 74% for women
- Motivation for the study: expose a threat to the privacy and safety of gay men and women

Let's discuss...

- Research question
 - Identification of sexual orientation from facial features
- Data collection
 - Photos downloaded from a popular American dating website
 - 35,326 pictures of 14,776 people, all white, with gay and straight, male and female, all represented evenly
- Method
 - A deep learning model was used to extract facial features + grooming features; then a logistic regression classifier was applied for classification
- Accuracy
 - 81% for men, 74% for women

What went wrong?

Questioning the ethics of the research question

- Identification of sexual orientation from facial features



Sexual orientation classifier - who can be harmed?

- In many countries being gay person is prosecutable (by law or by society) and in some places there is even death penalty for it
- It might affect people's employment; family relationships; health care opportunities;
- Gender, race, sexual orientation, religion are personal attributes and/or social constructs. Many of them can change over time. They can be non-binary. They are private, intimate, often not visible publicly.
- Importantly, these are properties for which people are often discriminated against.

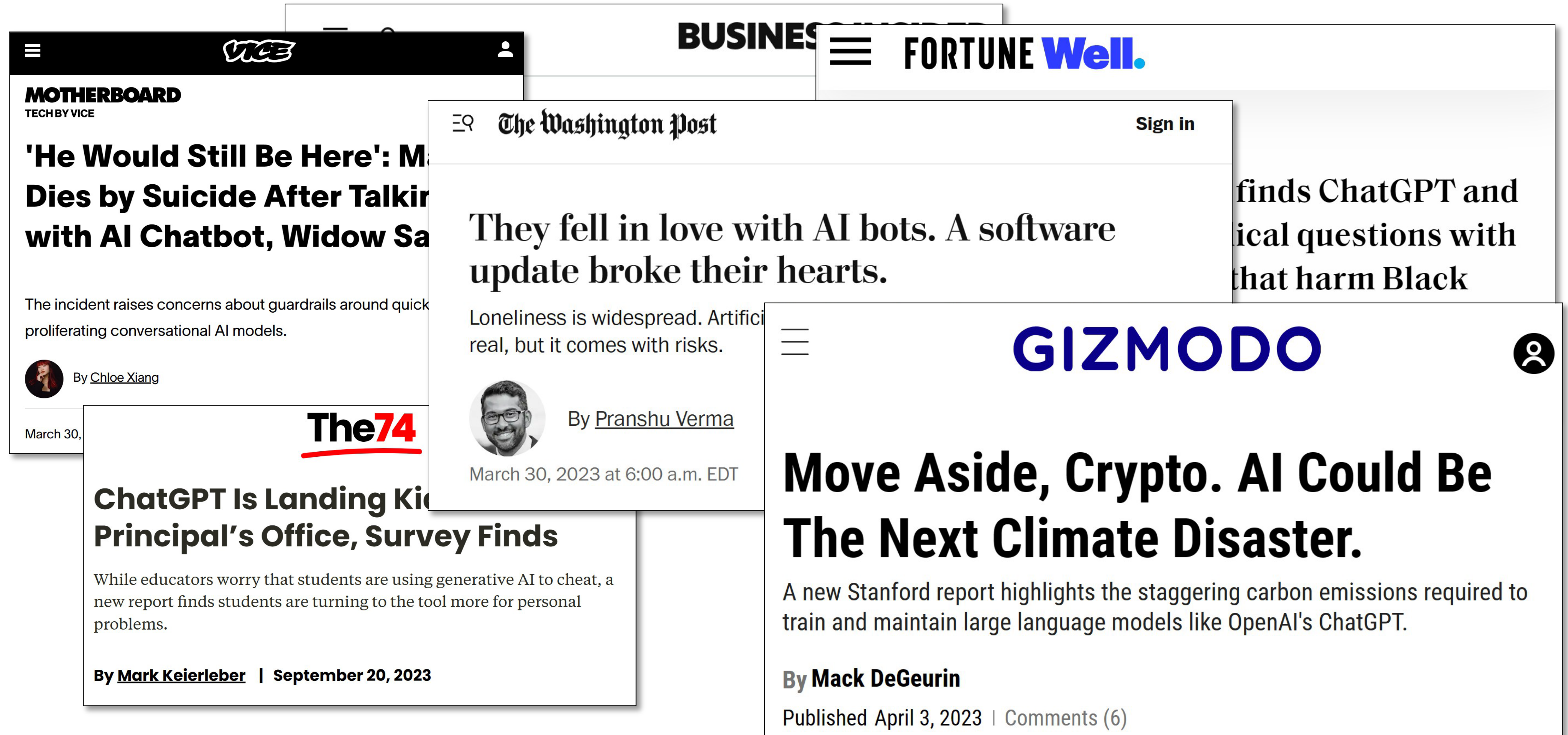
DALLE-2 result in 2022

- Image of a teacher:



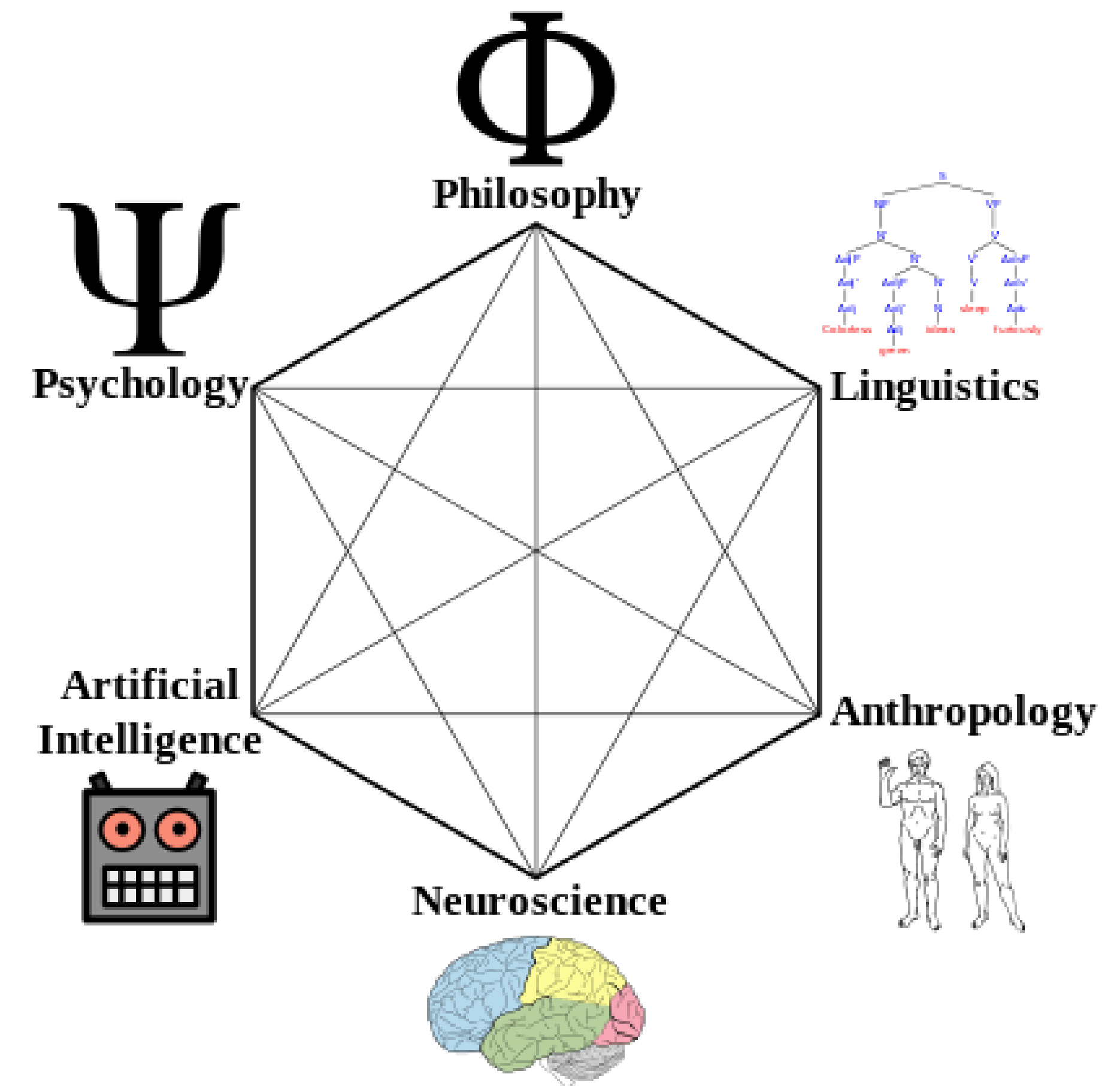
- *Let's discuss*: what do you notice about this image generation result?
 - Only women
 - Only white people
 - Certain age range

NLP tools have biases, and pose ethical risks



Ethics in AI

- Data origin and ownership
- Social bias in AI
- Algorithmic fairness
- Privacy risks and protection
- Mis-use of AI:
 - disinformation, opinion manipulation
- AI for good
 - content moderation
 - assistive technologies, disaster response
- Societal impacts of AI
 - environmental
 - economic, educational, policy impacts
-



Risks of Harms of NLP Models

		Theme	Subcategory
		Representational Harms	Stereotyping Demeaning Social Groups
Classification	Harm		
Discrimination, Exclusion and Toxicity	Social stereotypes and unfair discrimination Exclusionary norms Toxic language Lower performance for some languages and social groups		
Information Hazards	Compromising privacy by leaking private information Compromising privacy by correctly inferring private information Risks from leaking or correctly inferring sensitive information		
Misinformation Harms	Disseminating false or misleading information Causing material harm by disseminating false or poor information e.g. in medicine or law Leading users to perform unethical or illegal actions		
Malicious Uses	Making disinformation cheaper and more effective Facilitating fraud, scams and more targeted manipulation Assisting code generation for cyber attacks, weapons, or malicious use Illegitimate surveillance and censorship		
Human-Computer Interaction Harms	Anthropomorphising systems can lead to overreliance or unsafe use Creating avenues for exploiting user trust, nudging or manipulation Promoting harmful stereotypes by implying gender or ethnic identity		
Automation, access, and environmental harms	Environmental harms from operating LMs Increasing inequality and negative effects on job quality Undermining creative economies Disparate access to benefits due to hardware, software, skill constraints		

What we will **not** discuss

- . We will not discuss on notions of existential AI risk, and other related theories.
- . We will only discuss technical solutions, we will not discuss policy related solutions for AI safety.

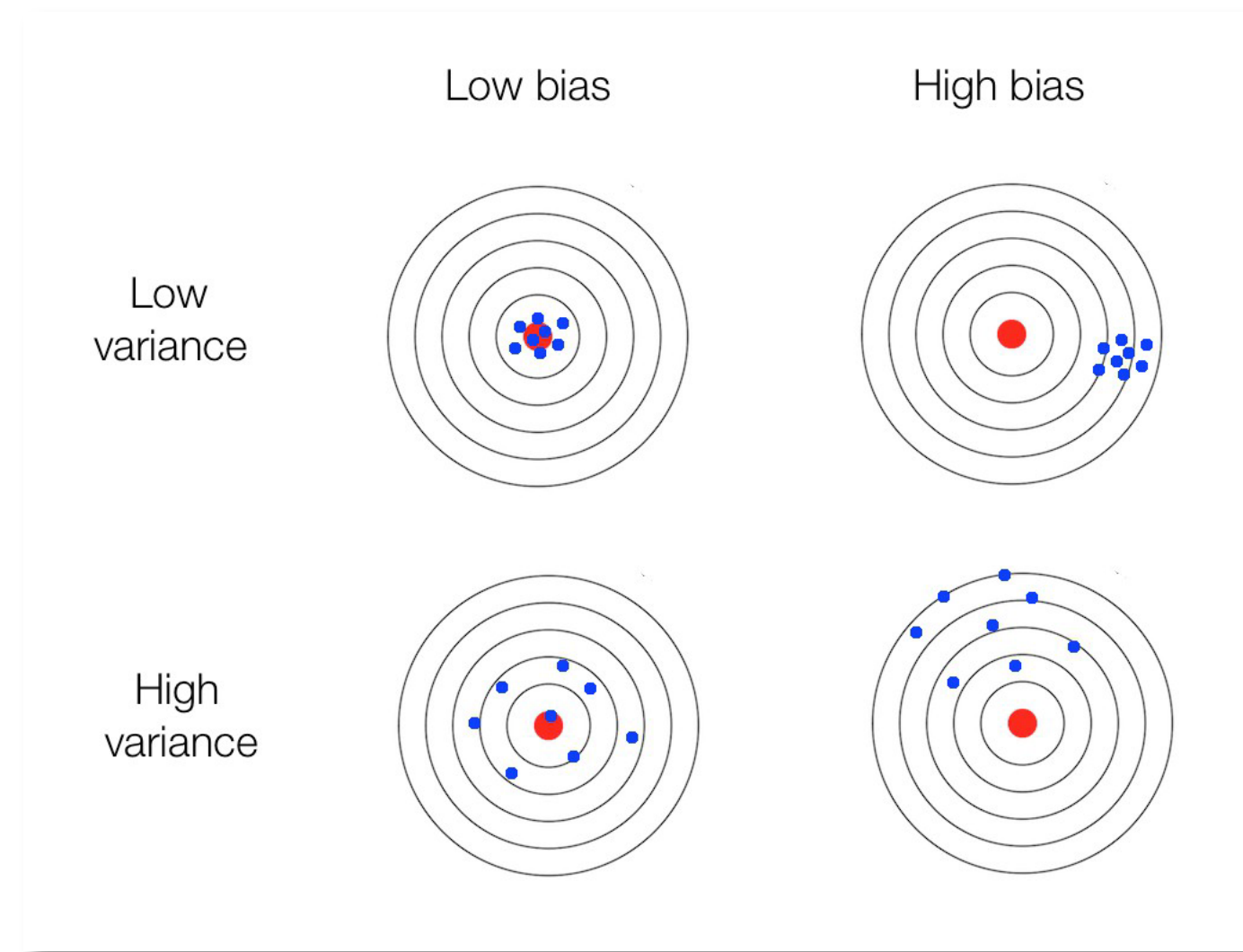
Today: a story in two parts



Part 1 – Bias

Some definitions of bias

- Bias [*statistics*]: systematic tendency causing differences between model estimates / predictions



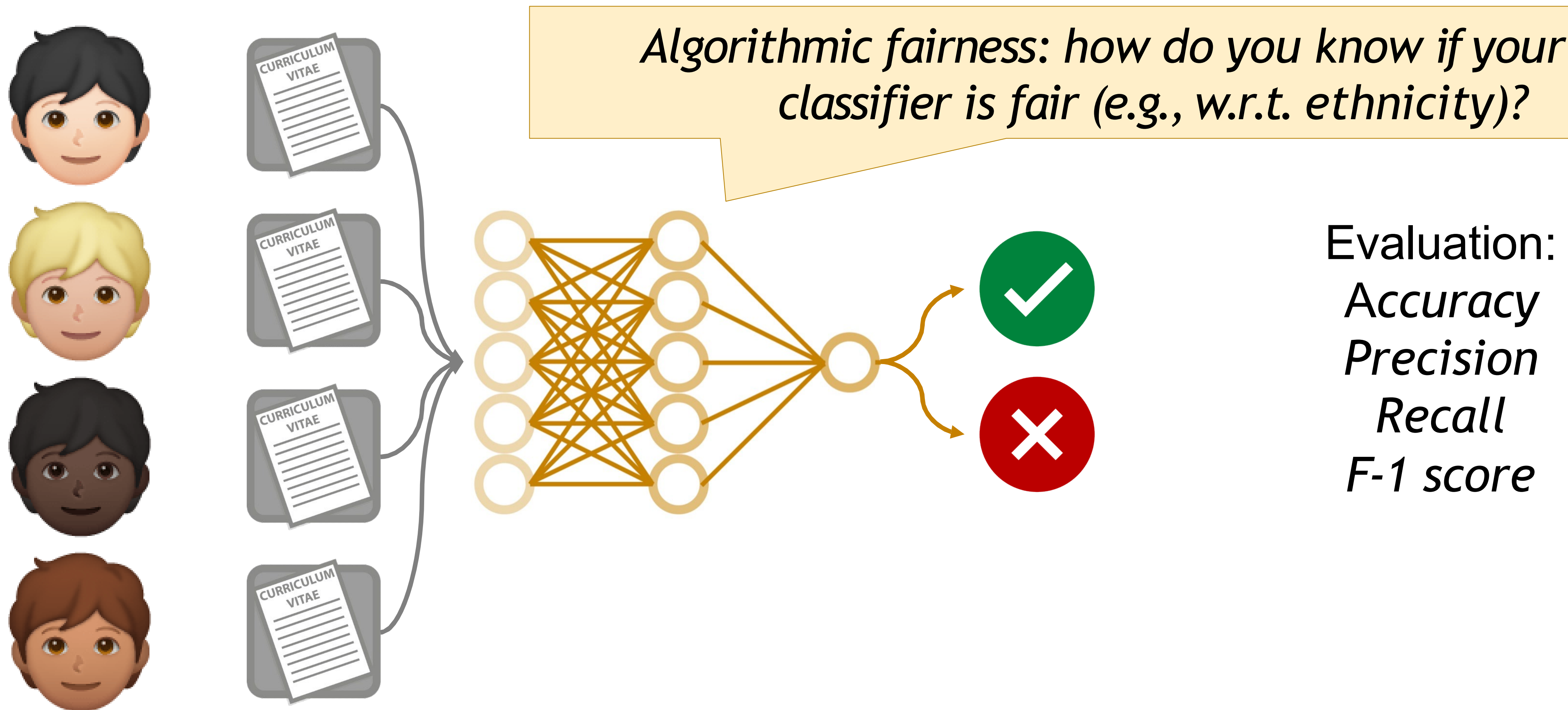
- Bias [*general*]: “disproportionate weight in favor of or against an idea or thing, usually in a way that is closed-minded, prejudicial, or unfair” – Wikipedia



Presence of bias $\simeq$ absence of fairness
Algorithmic fairness: attempts to correct biases in ML systems
But... how is fairness defined?

Algorithmic fairness

Let's assume a toy task: given a resumé, predict whether a candidate is qualified



Fairness metrics

- Accuracy quality: a classifier is fair if the people from different groups have the same accuracy
- Statistical parity: groups should have the same probability of being assigned positive class

 *Accuracy*
=

 *Accuracy*
=

 *Accuracy*
=

 *Accuracy*

$p(\checkmark | \text{Asian man})$
=

$p(\checkmark | \text{White woman})$
=

$p(\checkmark | \text{Black woman})$
=

$p(\checkmark | \text{Hispanic man})$

Equalized odds criterion [\[Hardt et al '16\]](#)

A classifier c is fair if the *false positive (FP)* and *true positive (TP)* rates are the same for different groups

- False positives

$$p(c(\text{👦}) = \checkmark \mid l(\text{👦}) = \times) \\ =$$

$$p(c(\text{👦}) = \checkmark \mid l(\text{👦}) = \times)$$

- True positives

$$p(c(\text{👦}) = \checkmark \mid l(\text{👦}) = \checkmark) \\ =$$

$$p(c(\text{👦}) = \checkmark \mid l(\text{👦}) = \checkmark)$$

		Predicted class	
		Positive	Negative
Actual class	Positive	TP	FN
	Negative	FP	TN

Other fairness metrics

- *Treatment equality*
 - Ratio of false negatives and false positives should be the same for groups
- *Fairness through unawareness*
 - Models should not employ sensitive attributes when making decisions
- *Causality-based*
 - *Counterfactual fairness*: outcome of the classifier would not changed if the sensitive attribute (e.g., race) were the only thing changed
- Many more...
 - [https://en.wikipedia.org/wiki/Fairness_\(machine_learning\)](https://en.wikipedia.org/wiki/Fairness_(machine_learning))
 - <https://fairmlbook.org/>

FAIRNESS AND MACHINE LEARNING

Limitations and Opportunities

Solon Barocas, Moritz Hardt, Arvind Narayanan

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We explore what makes automated decision making a matter of normative concern, situated in bureaucratic decision making and its mechanical application of formalized rules.

3 [CLASSIFICATION](#) [PDF](#)

We introduce formal non-discrimination criteria in a decision-theoretic setting, establish their relationships, and illustrate their limitations.

4 [RELATIVE NOTIONS OF FAIRNESS](#) [PDF](#)

Other fairness metrics

- *Treatment equality*
 - Ratio of false negatives and false positives should be the same for groups
- *Fairness through unawareness*
 - Models should not employ sensitive attributes when

FAIRNESS AND MACHINE LEARNING

Limitations and Opportunities

Solon Barocas, Moritz Hardt, Arvind Narayanan

- But, do these definitions really matter if no harms are caused?
Many argue that unfairness/bias should be measured in terms of the harms that it causes

- Many more...
 - [https://en.wikipedia.org/wiki/Fairness_\(machine_learning\)](https://en.wikipedia.org/wiki/Fairness_(machine_learning))
 - <https://fairmlbook.org/>

We explore what makes automated decision making a matter of normative concern, situated in bureaucratic decision making and its mechanical application of formalized rules.

3 CLASSIFICATION

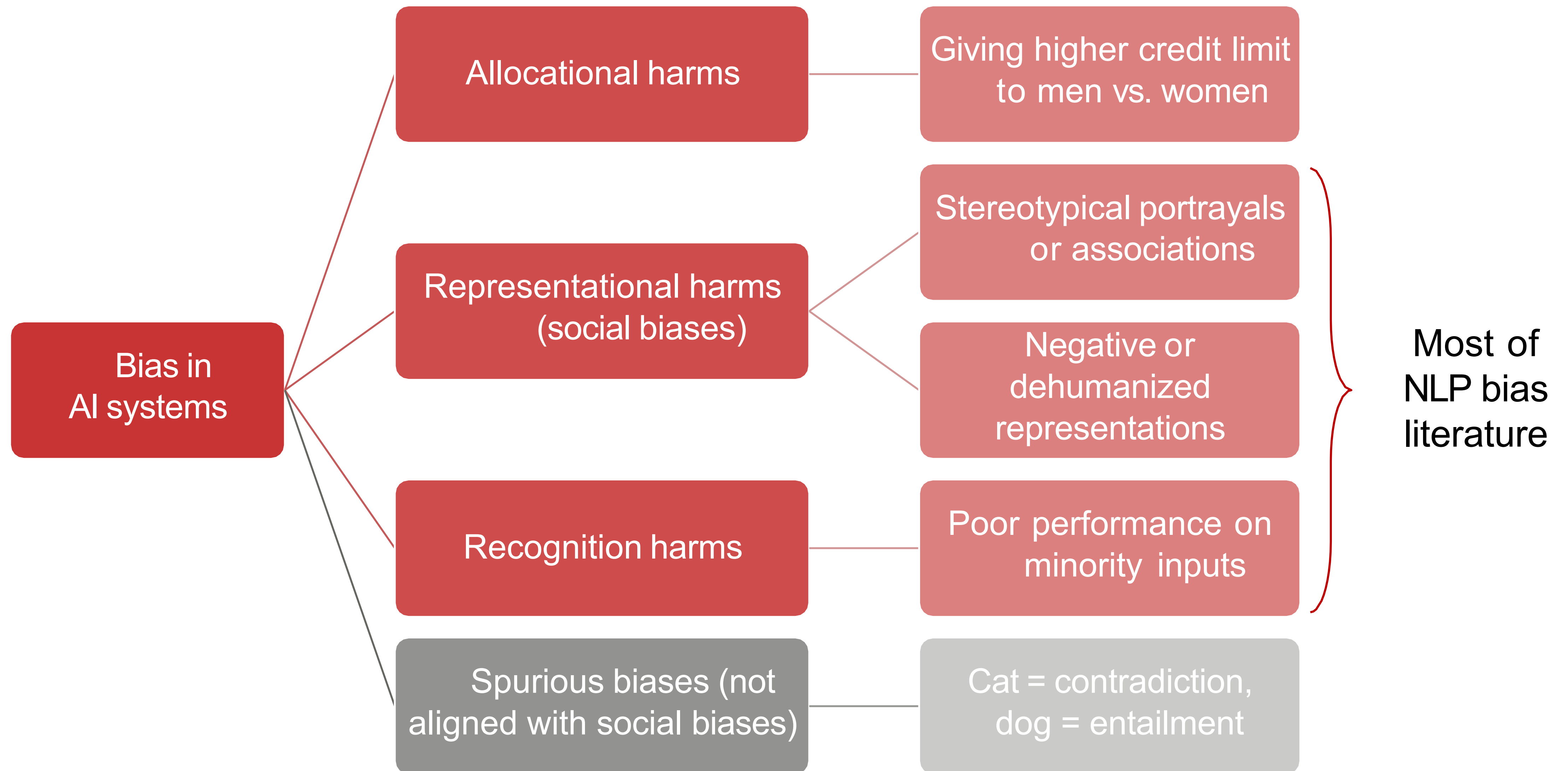
[PDF](#)

We introduce formal non-discrimination criteria in a decision-theoretic setting, establish their relationships, and illustrate their limitations.

4 RELATIVE NOTIONS OF FAIRNESS

[PDF](#)

Bias in terms of the harms it causes



“Bias” is an overloaded term

- [Blodgett et al 2020](#) examined ~150 NLP papers with “bias” in the title, found that many papers use term “bias” in ill-defined or vague ways

Some recommendations

Biased behavior

- What kinds of system behaviors are described as “bias”?
What are their potential sources (e.g., general assumptions, task definition, data)?

Harms from biases

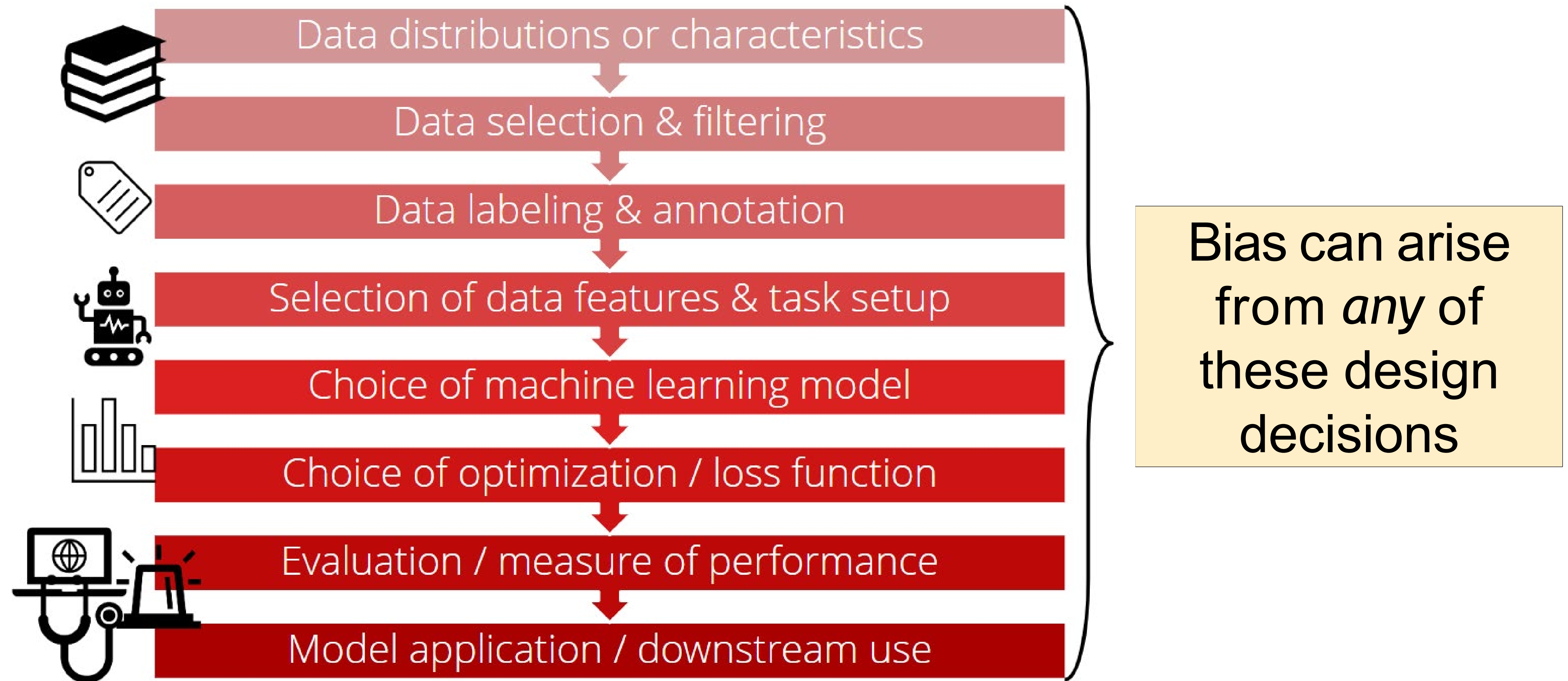
- In what ways are these system behaviors harmful, to whom are they harmful, and why?

Social values

- What are the social values (obvious or not) that underpin this conceptualization of “bias?”

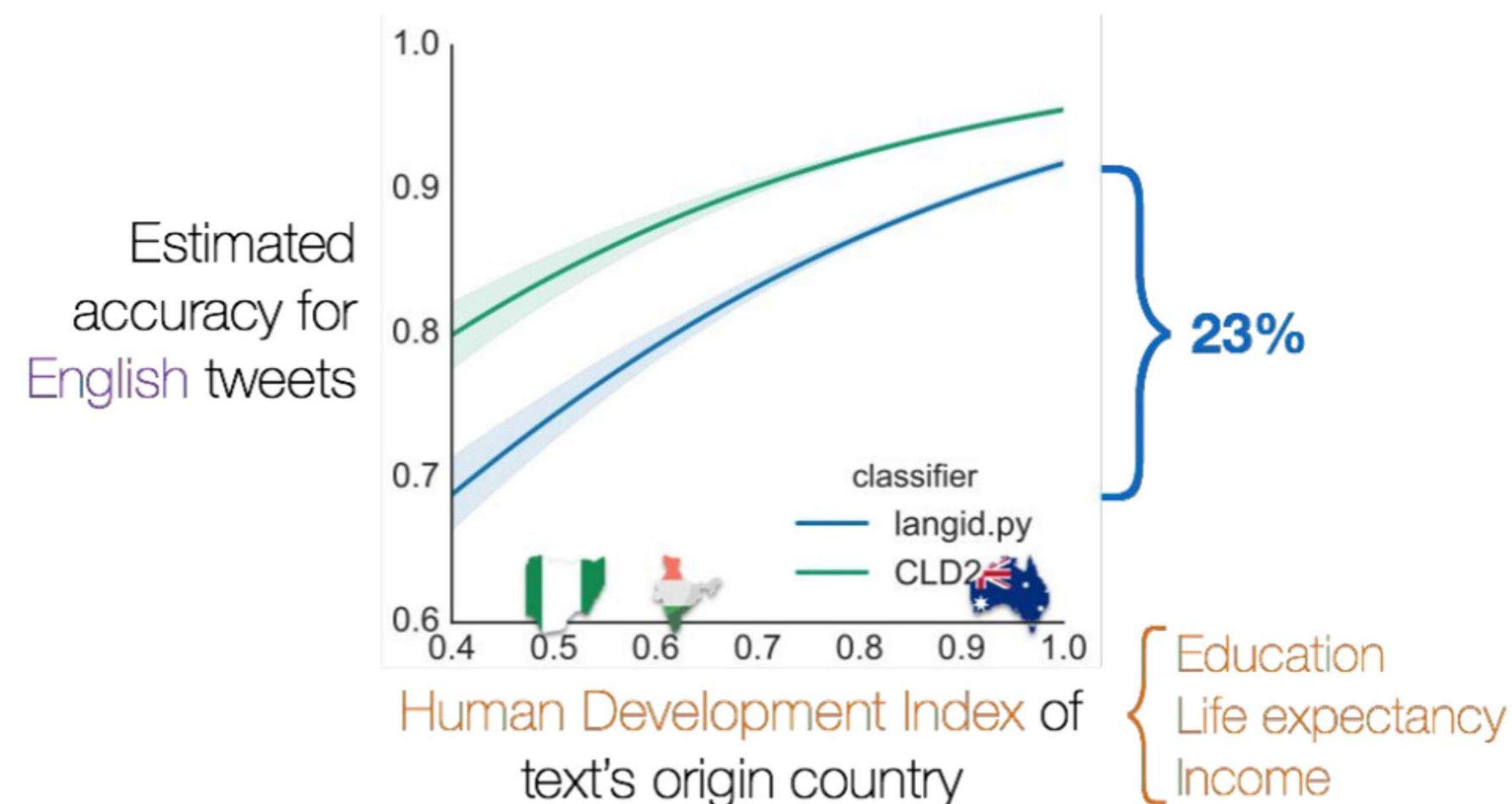
Where does bias come from?

Machine learning pipeline



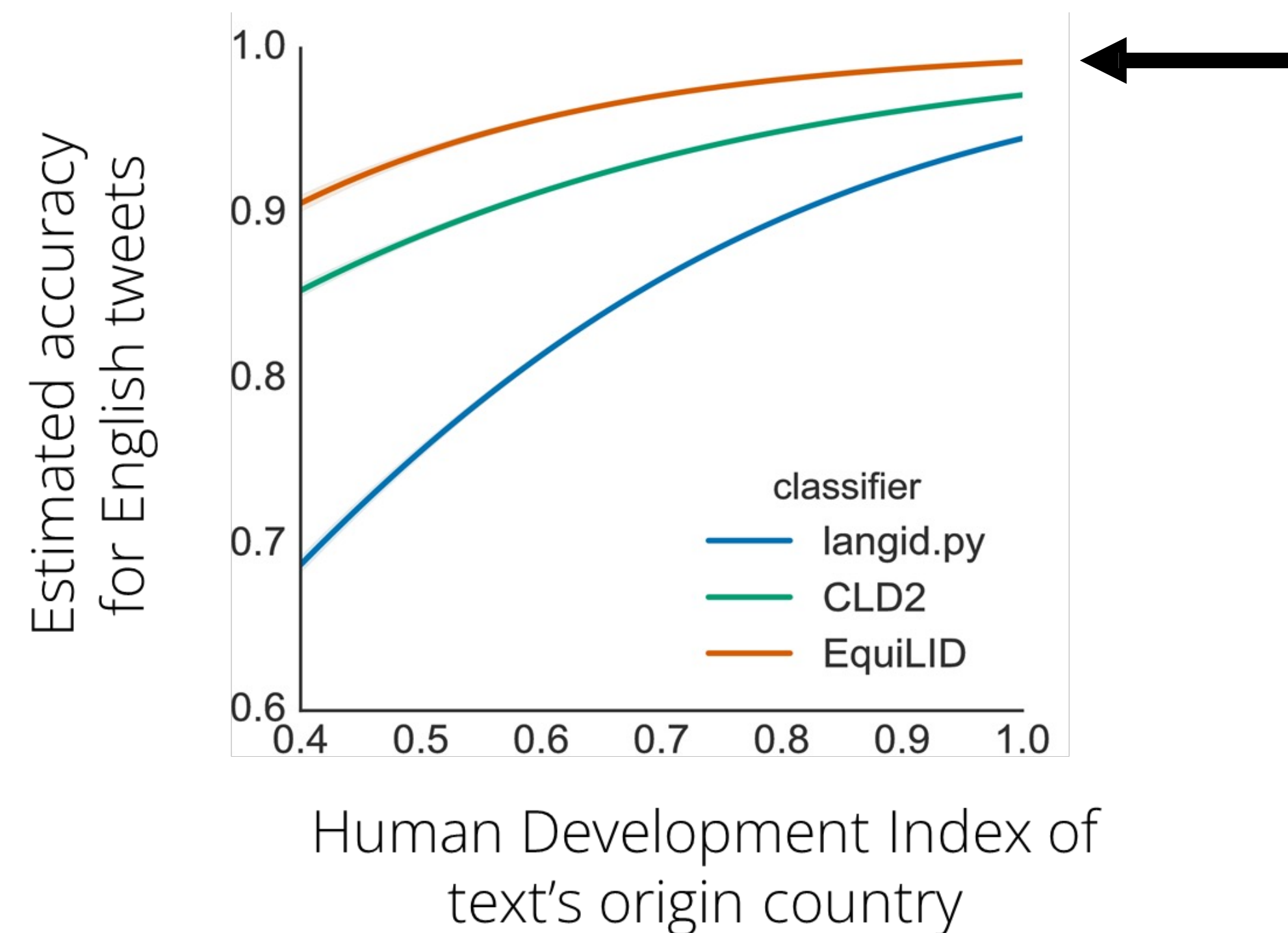
Example of bias from data: LangID tool

- LangID task: determine which language an input text is in
 - Considered a “*a solved problem* suitable for undergraduate instruction” (McNamee, 2005)
- Often a first step in most NLP and CSS preprocessing pipelines
 - e.g., filtering LLM pretraining data
- *But*, many variations of English in the world
 - *Int'l*: Nigerian English, Indian English, etc.
 - *Within US*: African American English, etc.
- [Jurgens et al. \(2017\)](#) found that accuracy of LID tool correlated with wealth/development level of country; works worse for low HDI countries

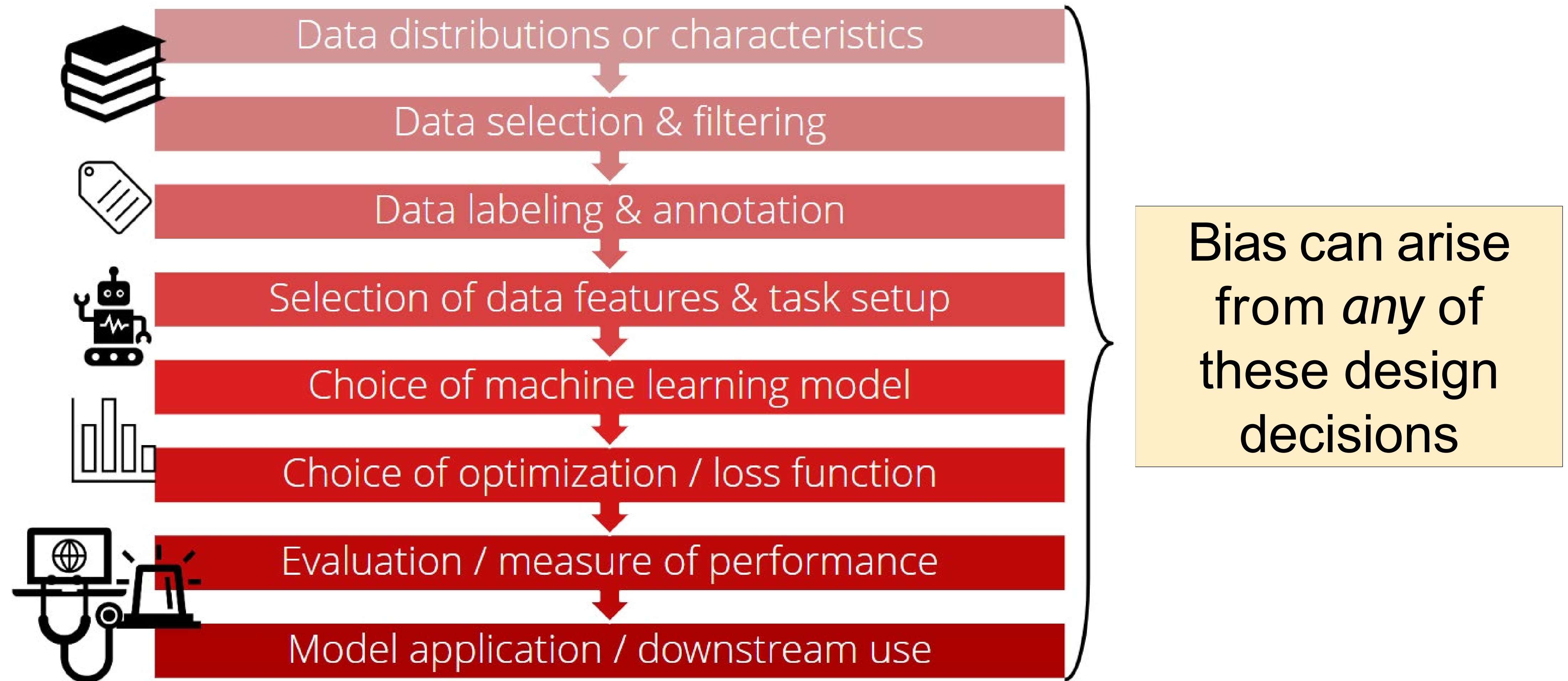


Example of bias from data: LangID tool (2)

- Jurgens et al (2017) introduce EquiLID
 - Trained by sampling more variety of data, topically, socially, geographically diverse, and even multilingual data
- Find that tool works much better than original LID systems
 - Bonus: even improved accuracy on highly developed countries!
- Takeaway: bias can be mitigated by making better data choices
 - But that's not the only source of bias...



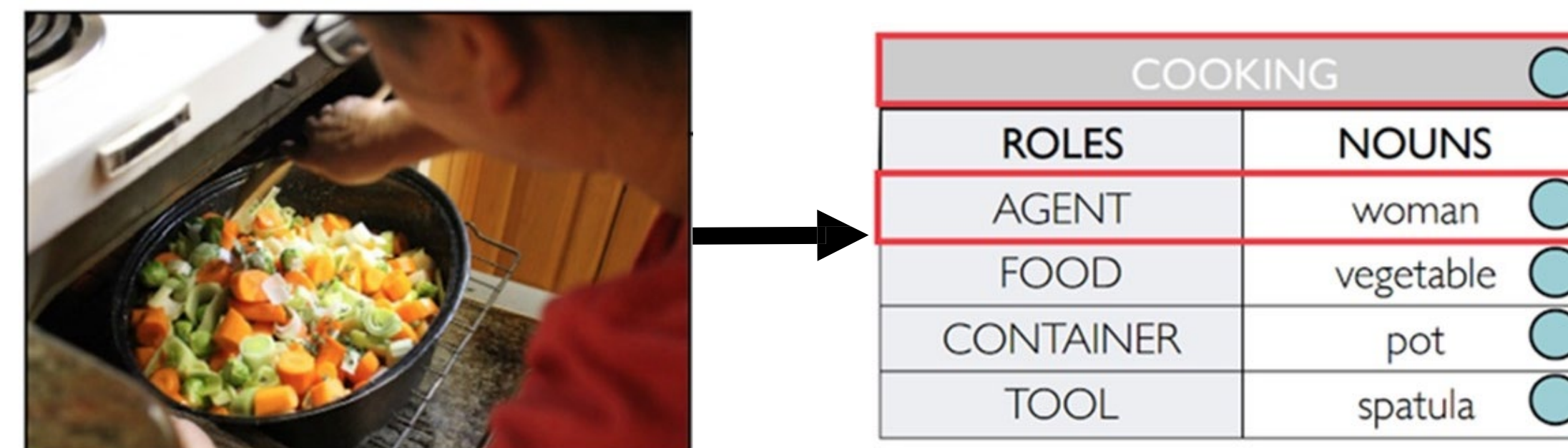
Machine learning pipeline



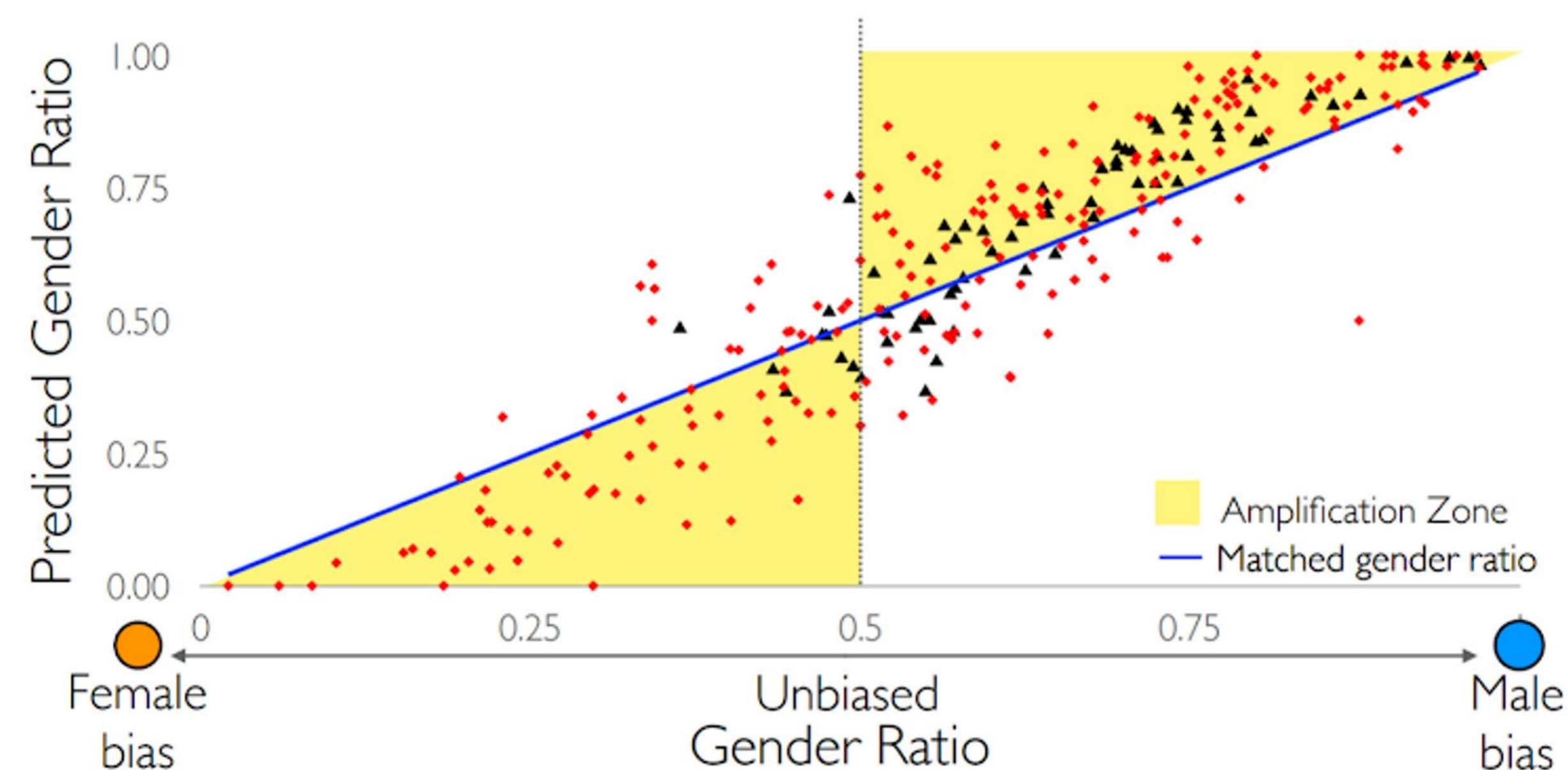
Bias amplification from models

21

- [Zhao et al \(2017\)](#) examined visual semantic role labeling task
 - Given an image, predict various semantic roles, including agent (person doing the action)
- Found skews in training dataset
 - E.g., 66% of training cooking images had agent=woman
- Found that models *amplified* biases
 - E.g., 84% of test cooking images predicted as agent=woman (~18% men mis-labeled)



◆ imSitu Verb
▲ COCO Noun



Model biases: mathematical links

- *Competing losses*: objective functions aim to minimize loss globally → learns to predict most frequent class
 - Often at the expense of less frequent classes (e.g. minority groups)
- *Simplicity bias*: neural networks biased towards learning simpler functions [[Valle Pérez et al. 2019](#)]
- Intuitively, if a model has limited learning capacity, makes sense that it learns shortcuts first
 - Shortcuts are often stereotypes or majority biases; e.g., CEOs are men
- Takeaway: ML/optimization choices also affect biases

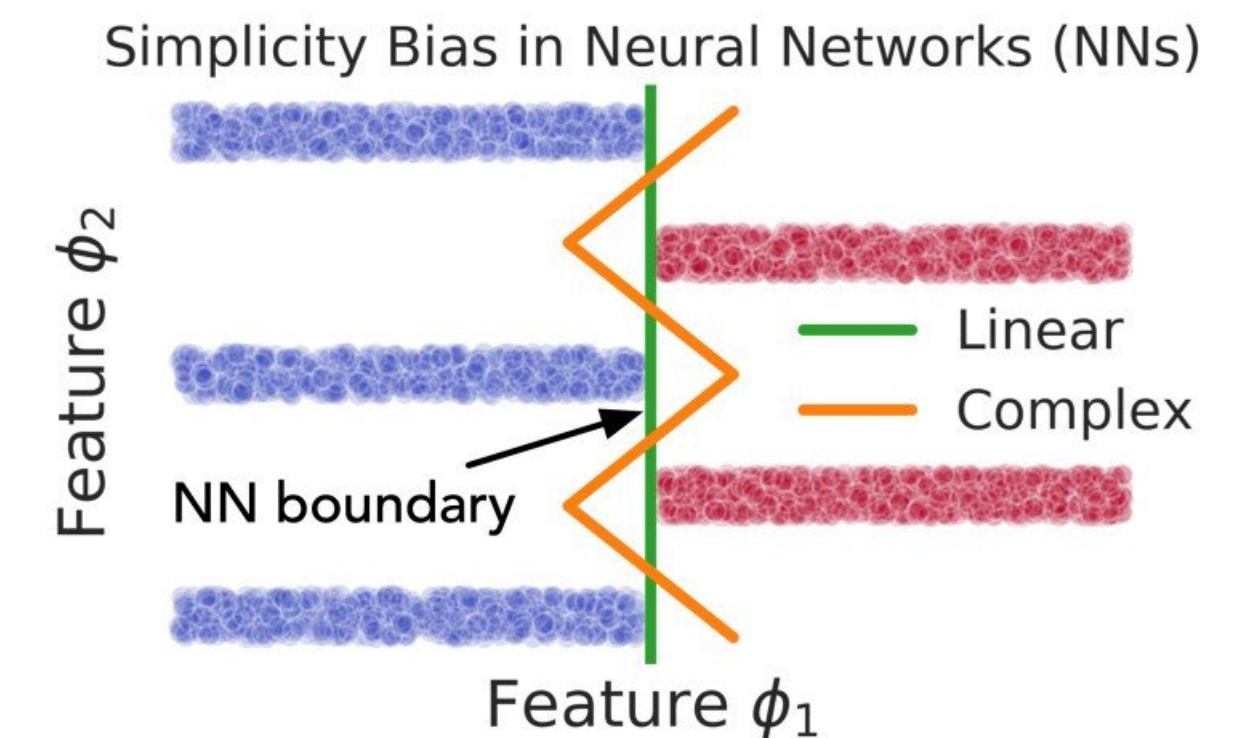
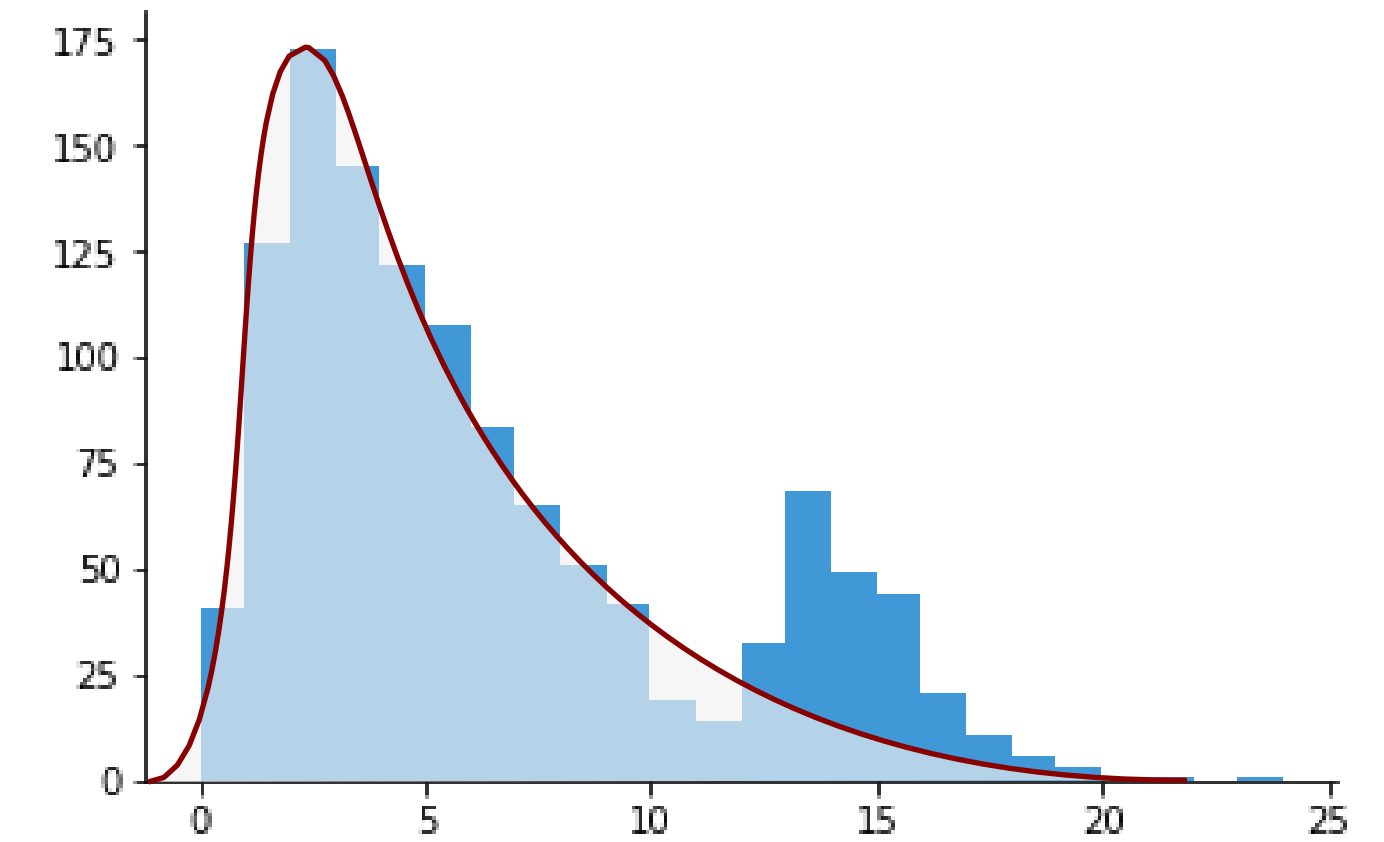
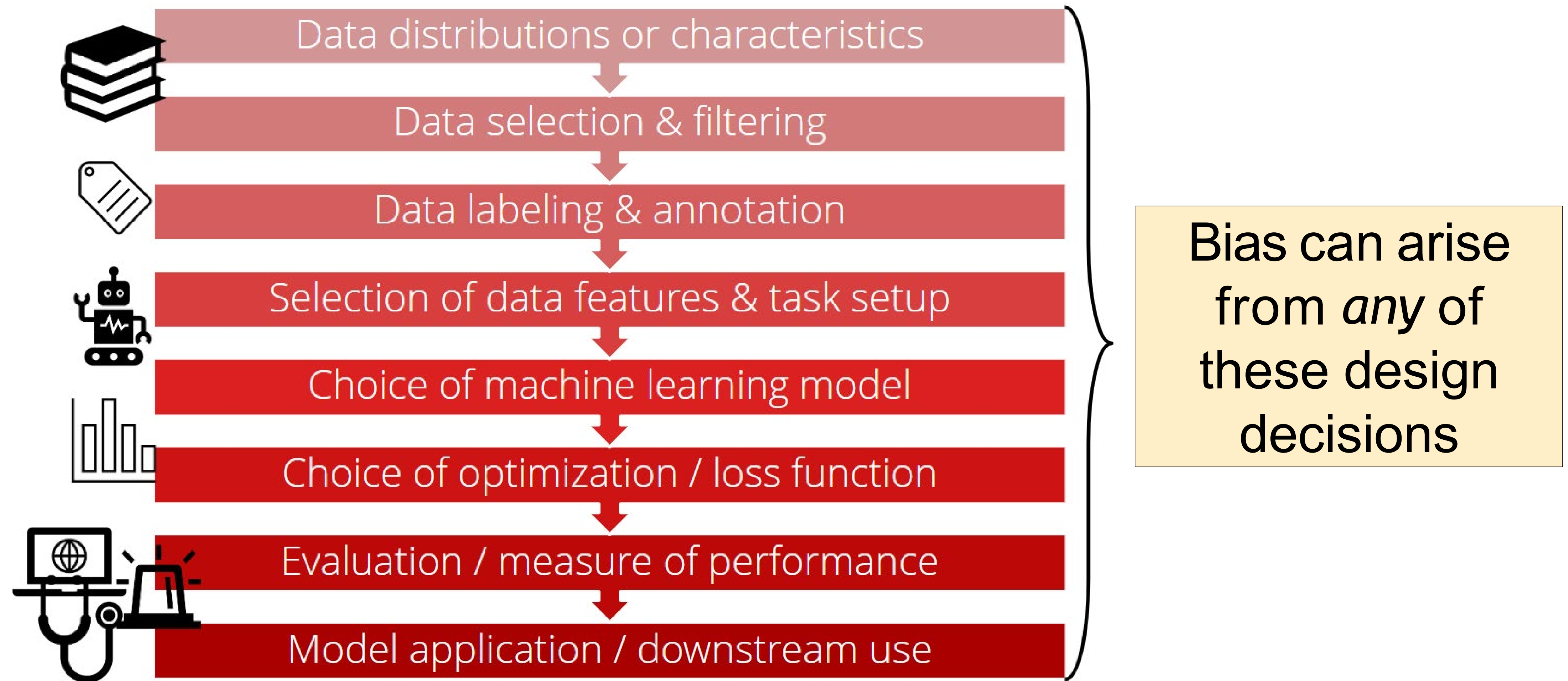


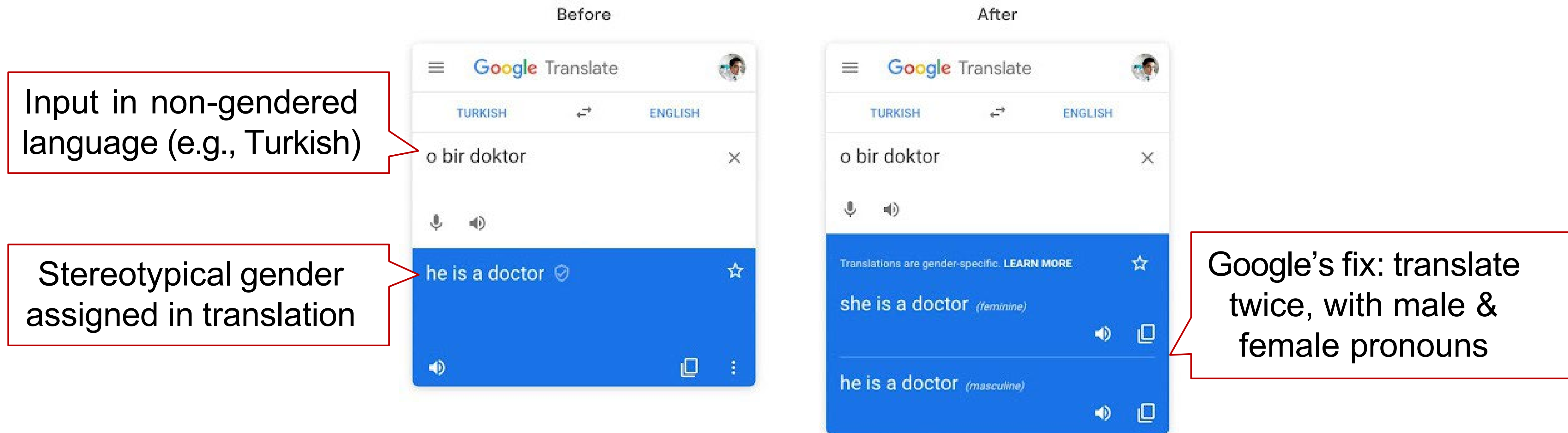
Figure 1: Simple vs. complex features

Figure from [Shah et al 2020](#)

Machine learning pipeline



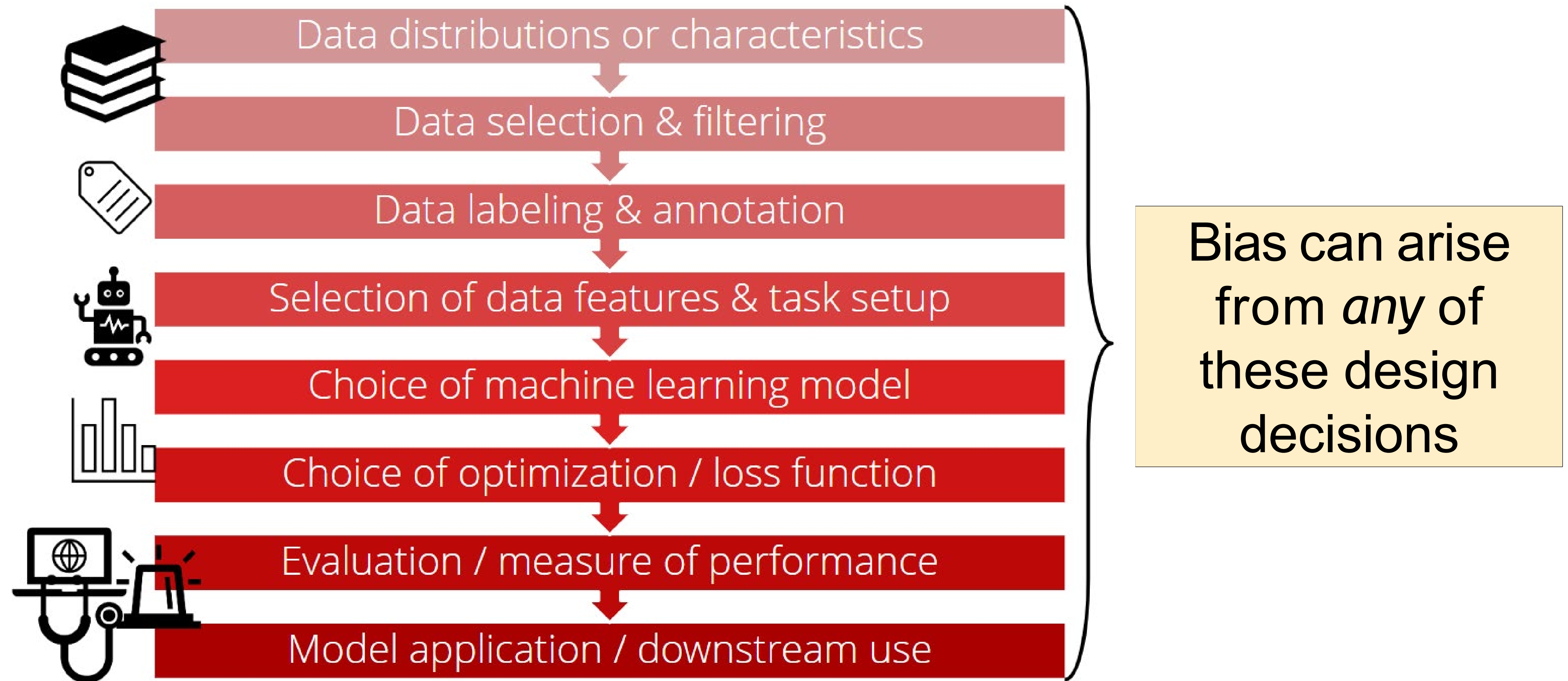
Google Translate issue



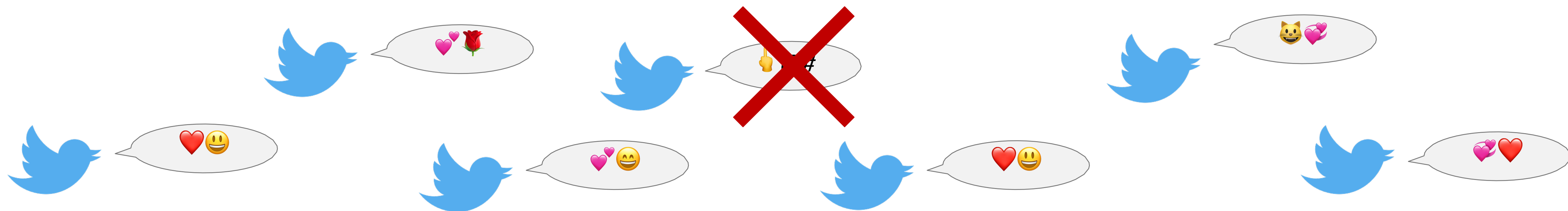
- Takeaways: mitigating bias may involve *system-level changes* to UI, input processing, output formatting, etc. while underlying AI model is similar
- *Let's discuss*: what do you think of this approach? What are some possible issues?

<https://ai.googleblog.com/2018/12/providing-gender-specific-translations.html>

Machine learning pipeline

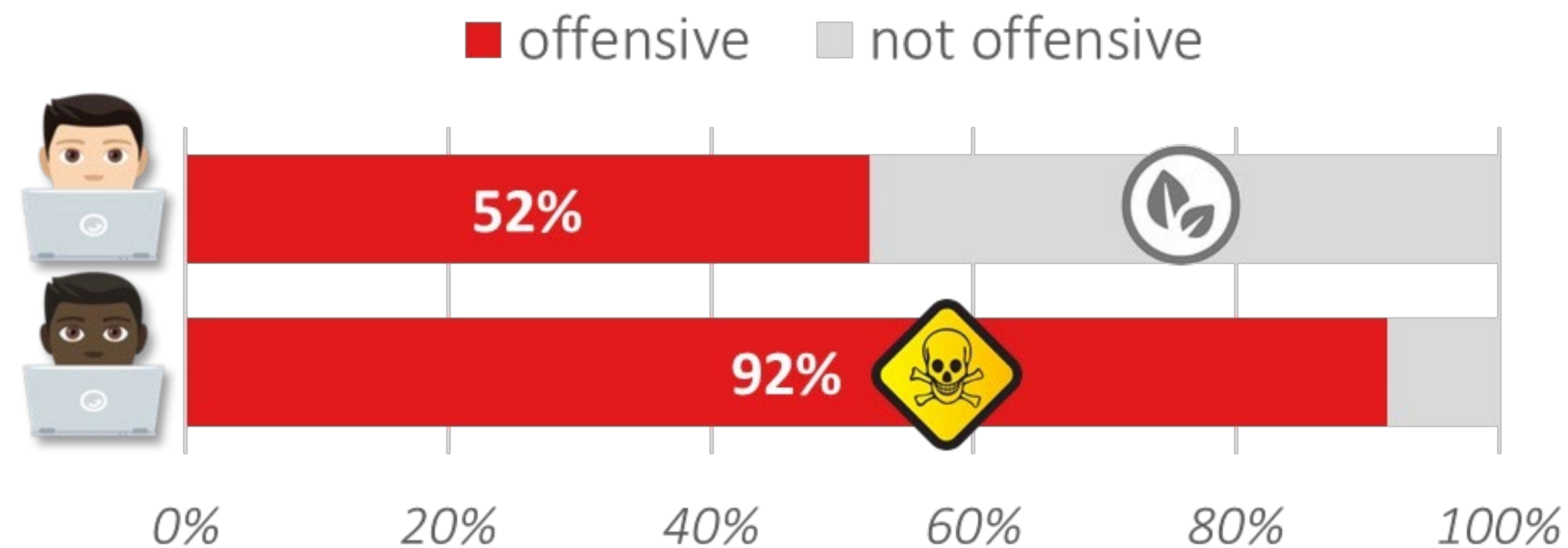


Hate Speech or Toxic Language Detection



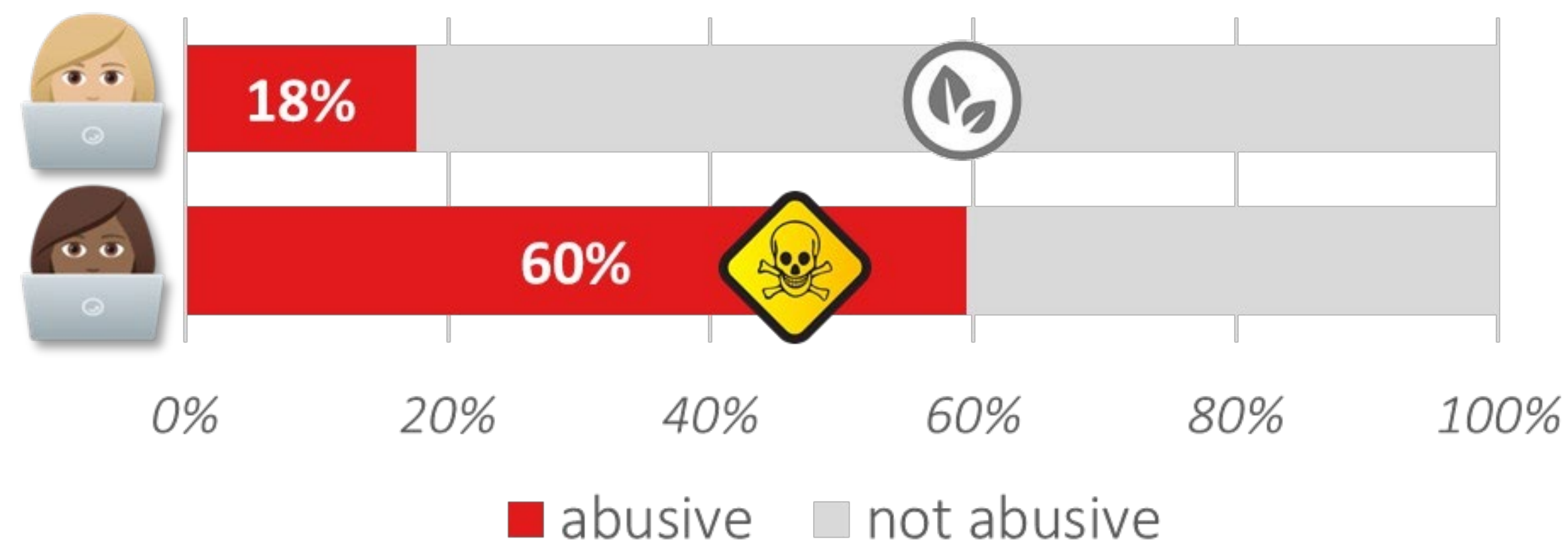
Goal: find and flag hateful or toxic content online, to make the internet less toxic

Racial biases in two popular datasets [Sap et al 2019]



TWT-HATEBASE
(Davidson et al., 2017)

Both datasets have **biases w.r.t. AAE tweets**



TWT-BOOTSTRAP
(Founta et al., 2018)

Enhancing the labeling interface [Sap et al 2019]

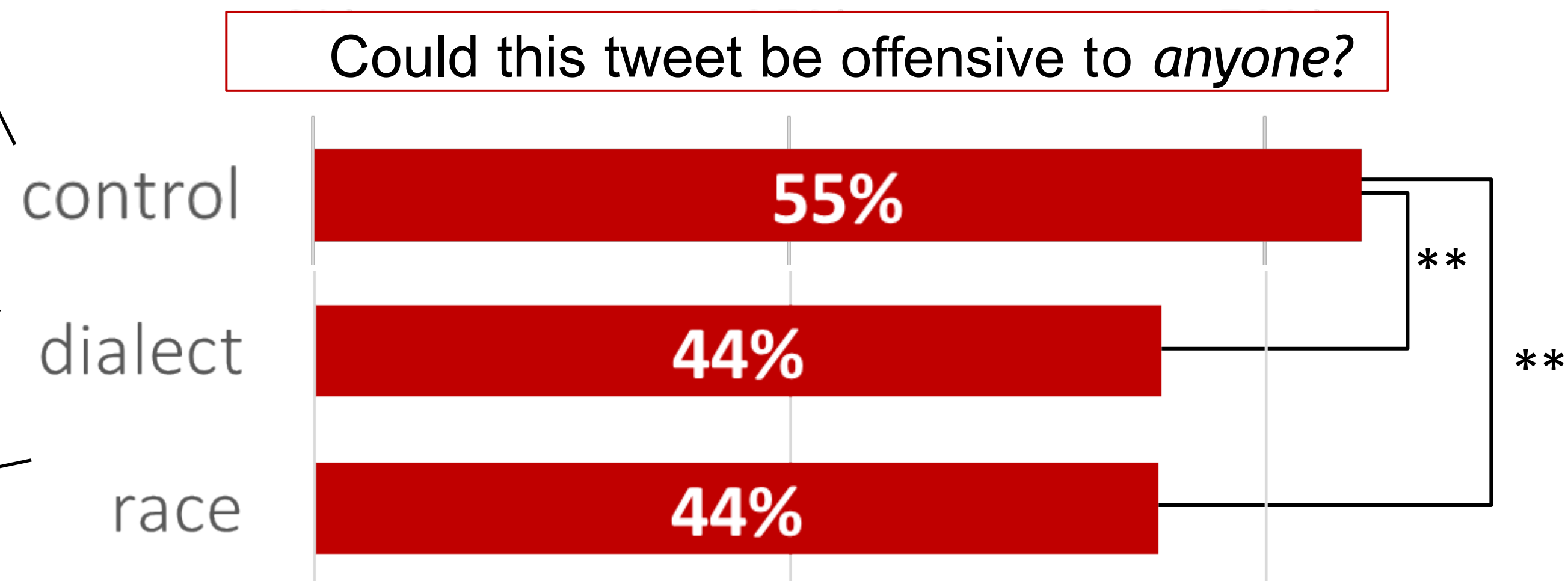
Control condition
Text-only, no context, prior work

Dialect priming
“Our AI thinks this tweet is in African American English”

Race priming
“A Twitter user that is likely Black/ African American tweeted...”

MTurk study:

- 350 AAE tweets, ~50% labeled toxic
- 3 (re-)annotators per tweet



Takeaway: adding social context to labeling mitigated bias

Why did these biases occur?

Why didn't NLP system designers think about these issues beforehand?

The world itself is biased



System designers have our own biases because of their *positionality*, i.e., set of perspectives that we hold due to our lived experiences and identity.

Positionality affects all our choices (e.g., assuming 1-1 mapping between languages and gendered pronouns, assuming toxicity looks the same in different dialects)

Debiasing AI systems

Is it even possible?

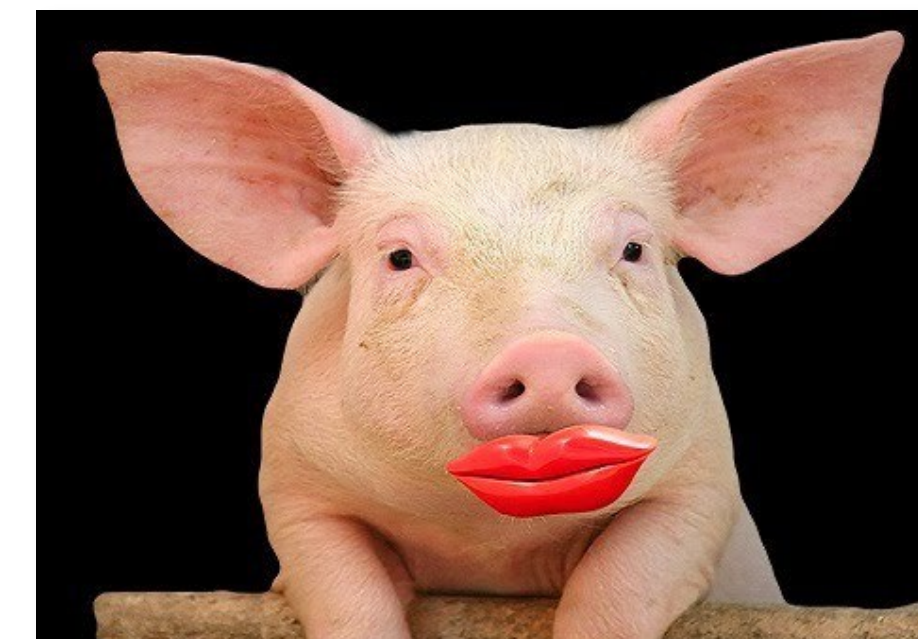
DALLE-2 vs. Gemini



- DALL-E-2 generated images were shown to have social biases, later fixed by adding identity keywords to the input prompts (e.g., prompt+“ *Asian*”; [Sparkes 2022](#))
 - Gemini generations also shown to have skews
- *Let's discuss*: what do you think of this? How are these two generations different?

Limits of debiasing

- Gender debiasing doesn't work
 - Breaks down for non-binary genders, racial categories or other social identity types
- Intrinsic debiasing $\neq$ actual debiasing
 - Finetuning often reintroduces biases
 - Out-of-distribution data often still show biases
- *Real world vs. ideal world*: is reflecting the (biased) status quo the goal? or do we want to build a more fair or just world?
- Justice and fairness go beyond data & model fairness



“Lipstick on a pig” paper,
Gonen & Goldberg 2019

On the Intrinsic and Extrinsic Fairness Evaluation Metrics for Contextualized Language Representations

Yang Trista Cao^{*1}, Yada Pruksachatkun^{*2}, Kai-Wei Chang^{2,3}, Rahul Gupta²
Varun Kumar², Jwala Dhamala², Aram Galstyan^{2,4}

¹University of Maryland, College Park



A lot of people have understood that we need to have more diverse datasets, but unfortunately, I felt like that's kind of where the understanding has stopped. It's like *'let's diversify our datasets. And that's kind of ethics and fairness, right?'* But you can't ignore social and structural problems.



Timnit Gebru, PhD

Socio-technical view on bias & fairness

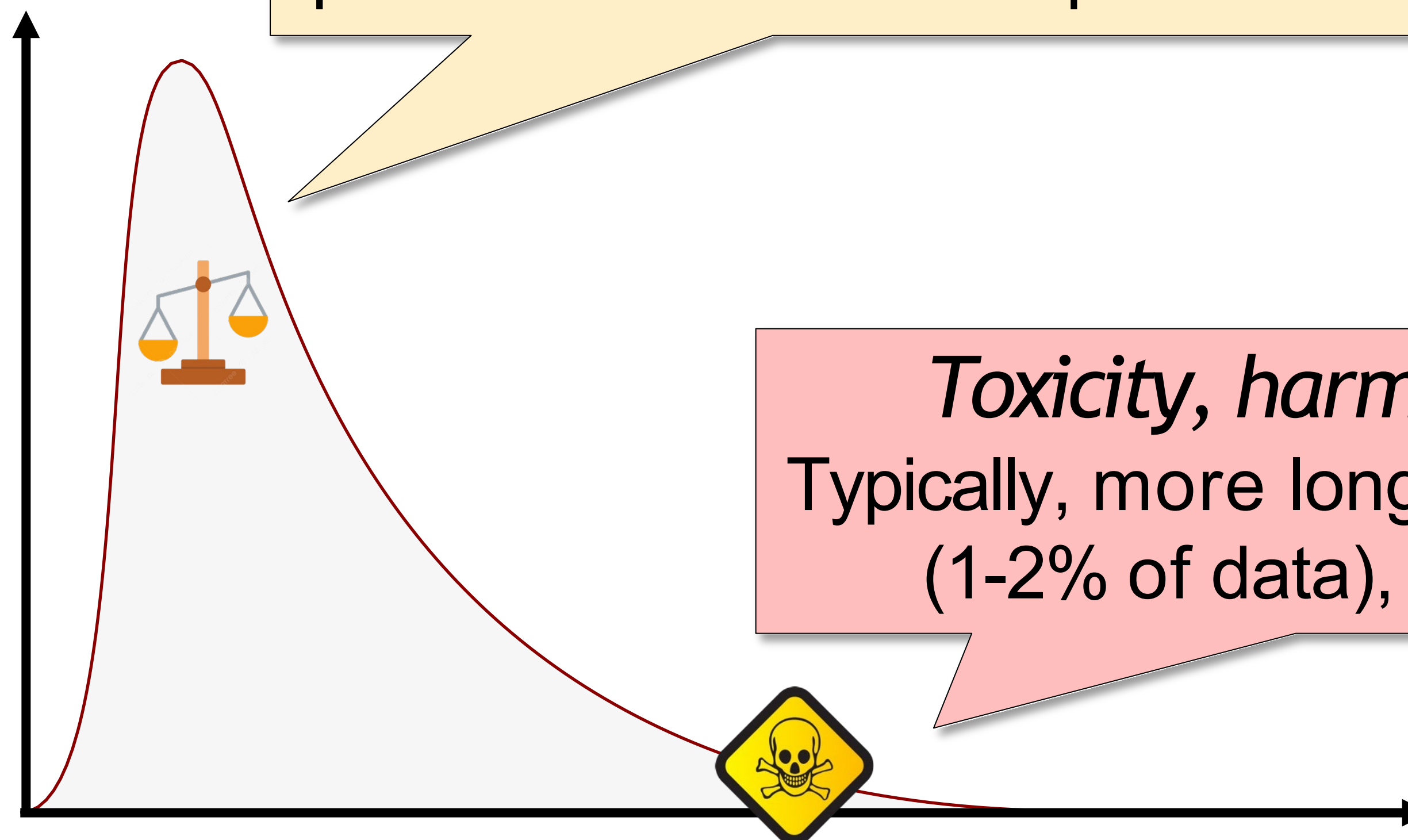
- You can have an “fair” NLP/ML model (e.g., facial recognition system)
 - 95% accuracy/error rate on white & Black faces
- But if the system is used by law enforcement, bias creeps in w.r.t. who the system is used on
 - Black people more often arrested, due to racial biases
- Actual error rates are a function of deployment
- Algorithm’s fairness $\neq$ fairness of treatment



Part 2: Harmful content & toxicity

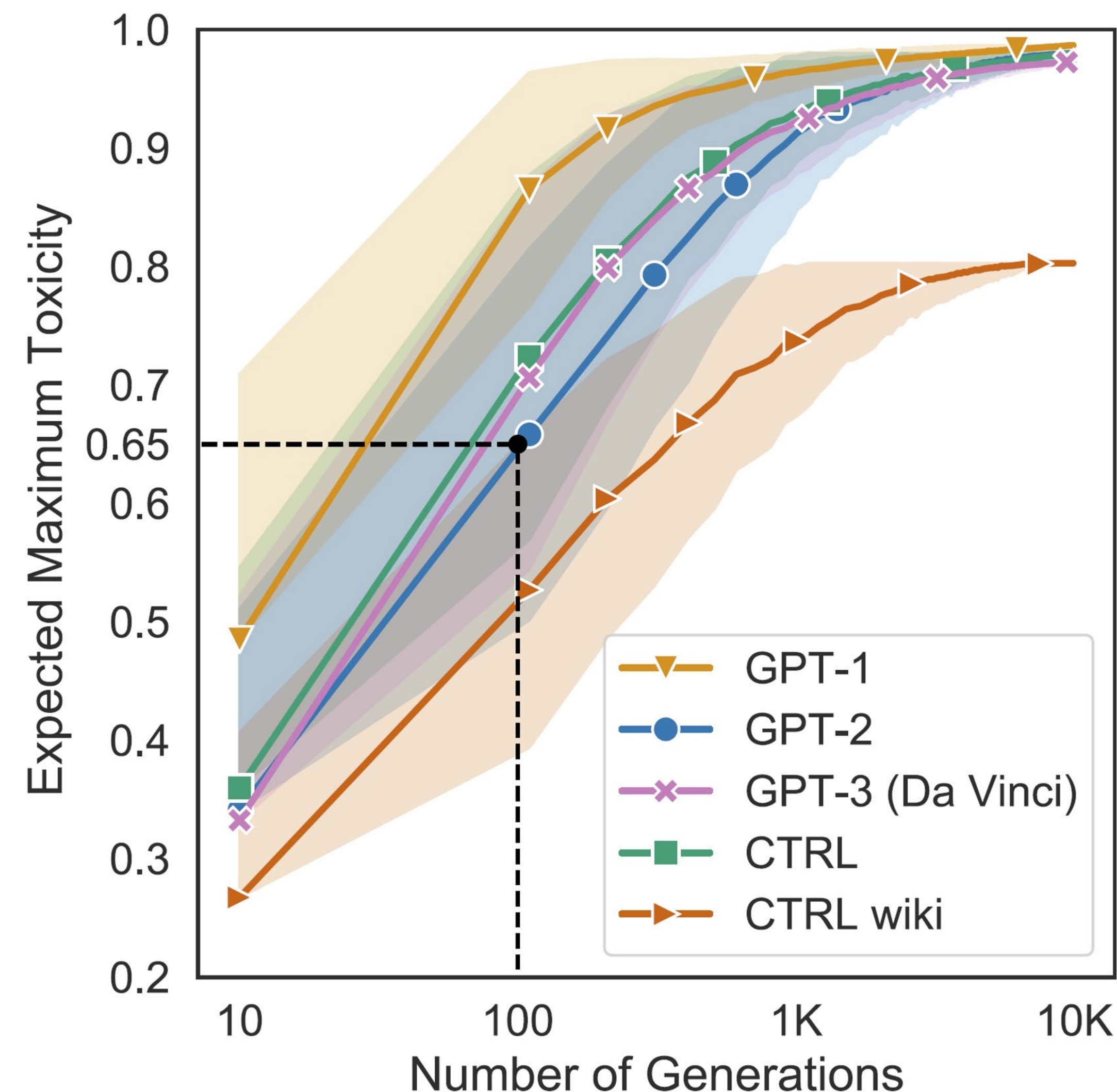
Biases vs. toxicity

Social biases & stereotypes:
Pervasive patterns that are prevalent in most data points



Toxicity in LLMs, how bad is the problem really?

- [Gehman et al \(2020\)](#) introduced concept of *neural toxic degeneration* in LLMs
- Out of a 100 generations sampled from models, at least one toxic sentence
 - 65-70% toxicity from GPT2, GPT3
 - 85% toxicity from GPT1
- Model size affects toxicity: larger models have more toxicity [[Touvron et al 2023](#)]



Why are these models learning so much undesirable content?

Problems with self-supervised pretraining

“Feeding AI systems on the world’s beauty, ugliness, and cruelty, but expecting it to reflect only the beauty is a fantasy”

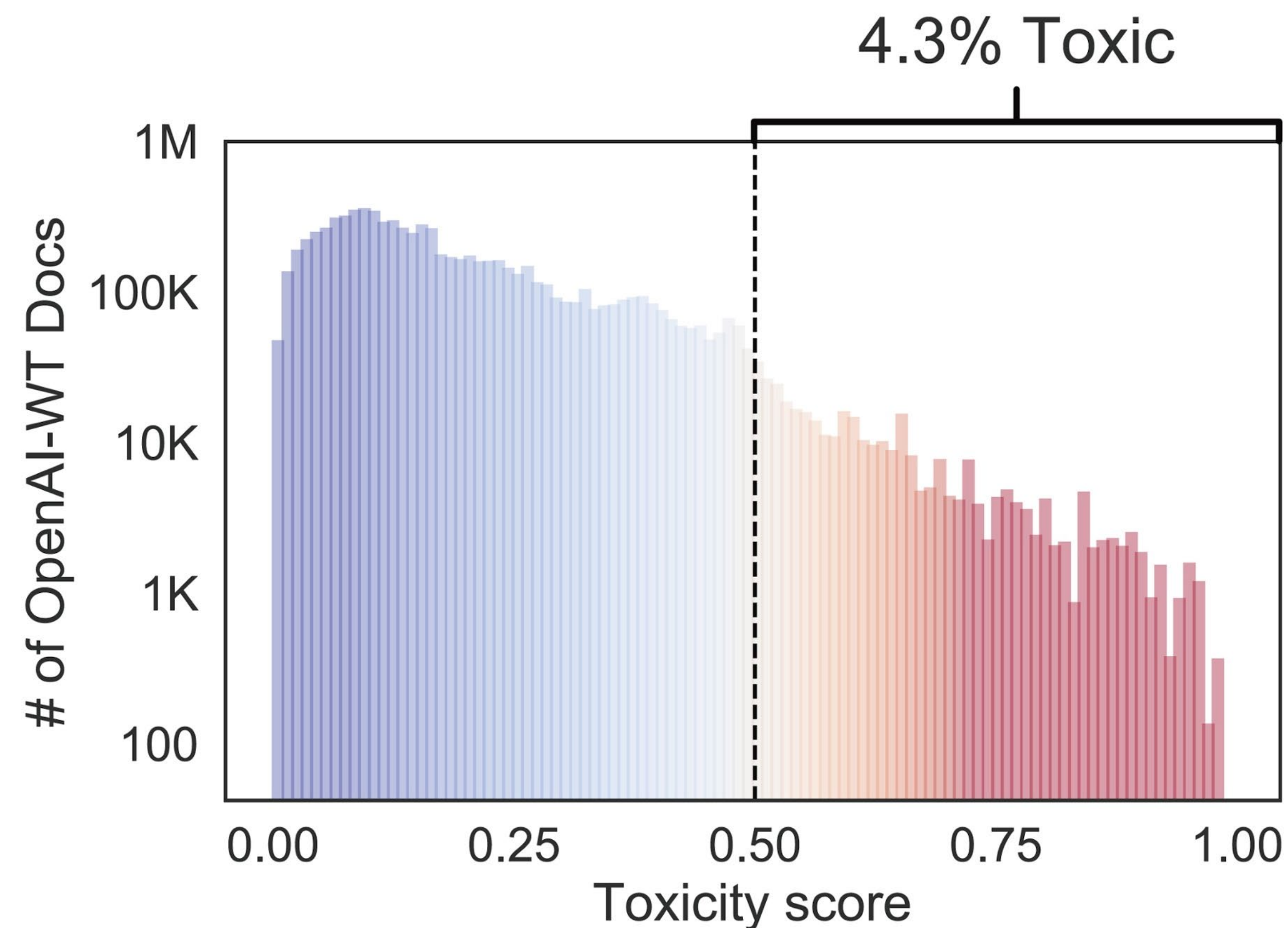


Prof. Ruha Benjamin, PhD

- Recipe: scrape as much pretraining data as you can to train your LM
- Consequence: LM ends up learning toxicity, biases, extremism, hate speech...

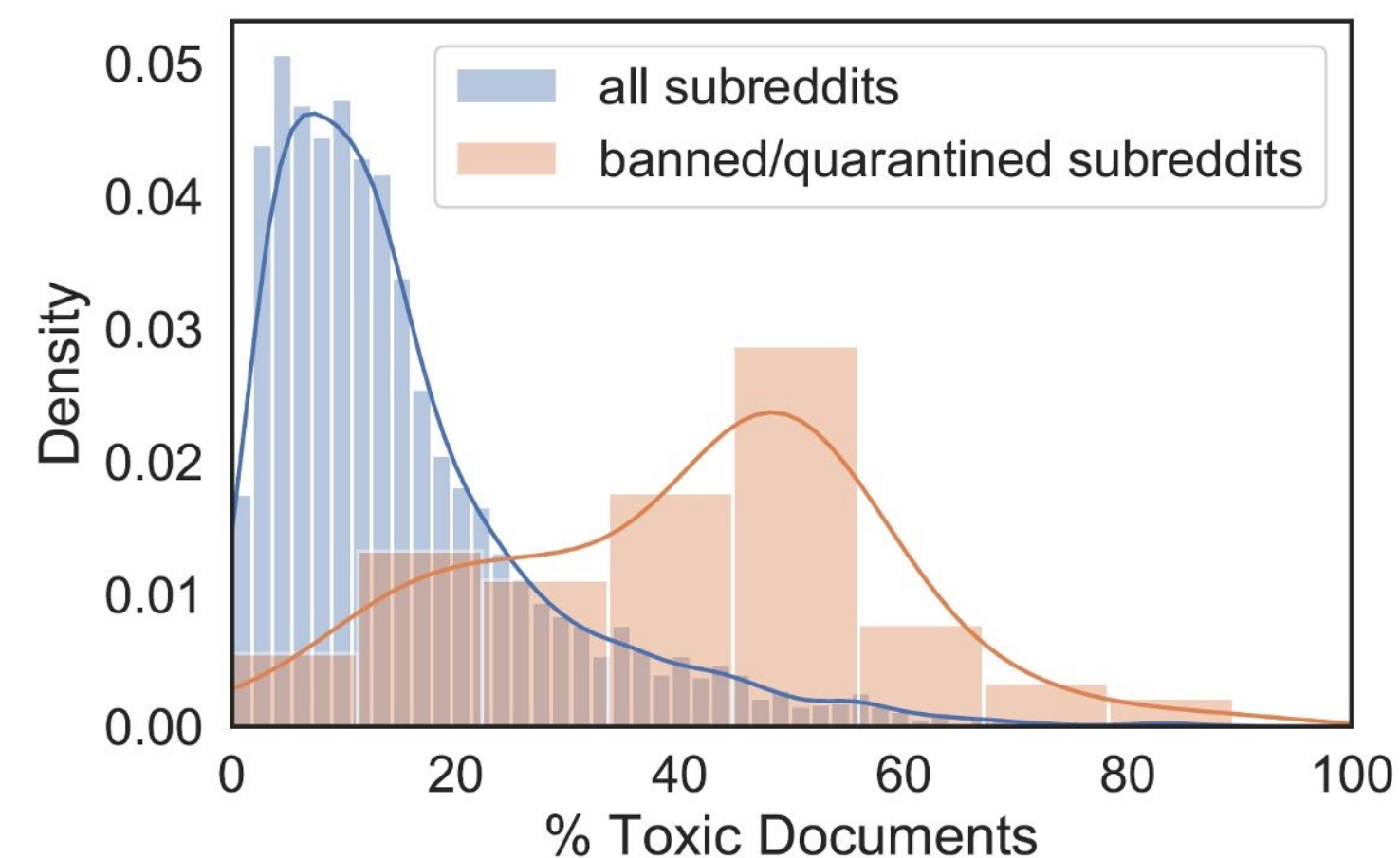
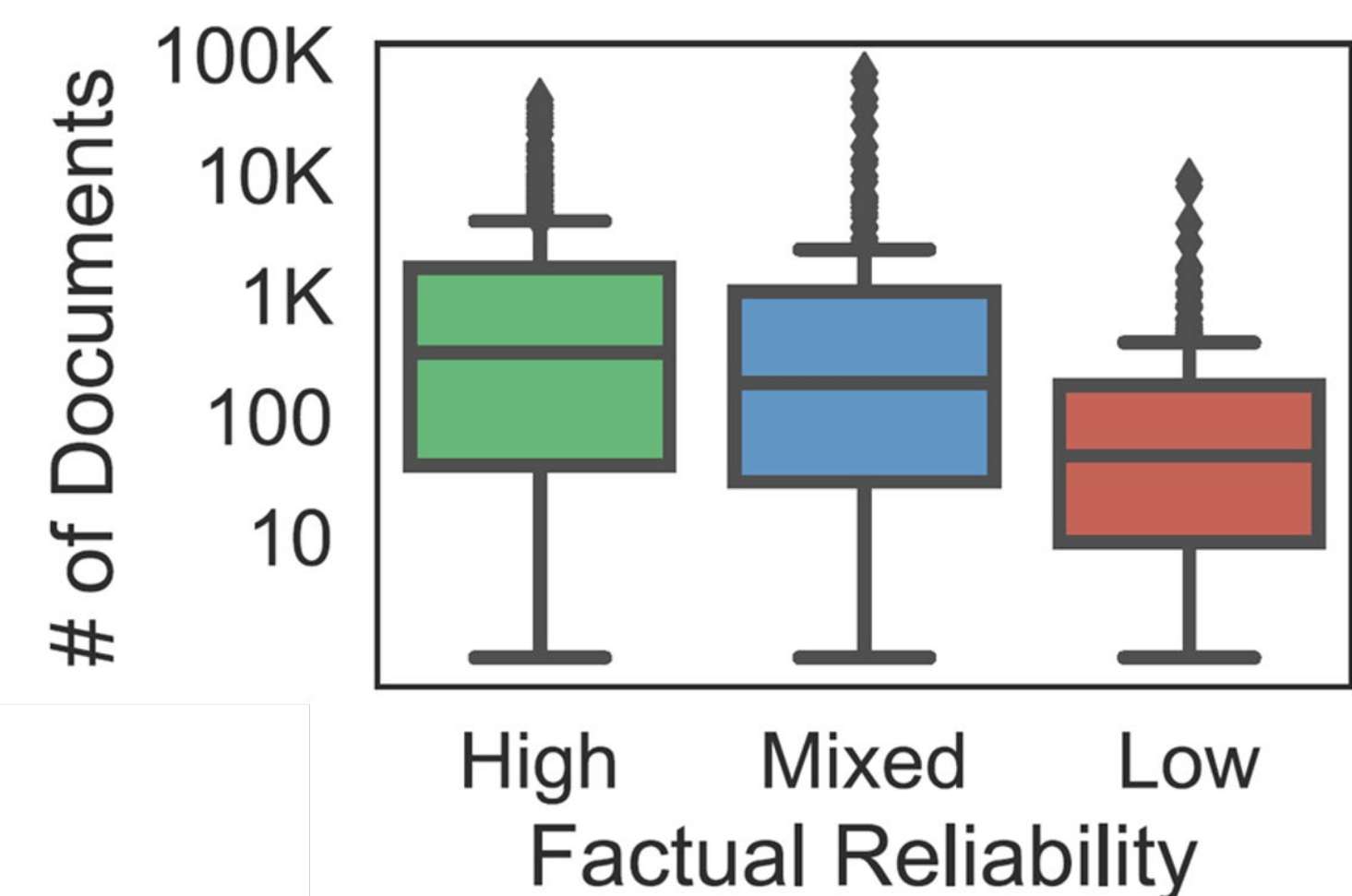
Toxicity in GPT-2's pretraining data

- [Gehman et al \(2020\)](#) accessed the actual GPT-2 training corpus (OpenAI-WT)
 - 8 million documents, 38Gb of text
 - Outbound links from Reddit posts with Karma ≥ 3
- Scored it with PerspectiveAPI toxicity
- Found >4% of documents (340,000) are toxic



Fake news in GPT-2's pretraining data

- Also looked at sources of documents in training data
- Cross-referencing sources of documents with known factual reliability categorization
 - >272K (3.4%) docs from low/mixed reliability sources
- Examining source where document is shared
 - >200K (3%) docs linked from banned/quarantined subreddits, which typically are more toxic docs
- Important to examine training data
 - Can only do that if publicly released!
- *So... need approaches to safeguard your model against this undesirable content, knowledge, and text.*



How to safeguard your LLMs

Overview – LLM safeguarding

Safeguards from training data

- Filtering out toxic training data

Safeguards from input prompt classification

- Topic-based filters
- Toxic content detection

Safeguards from instruction-tuning & RLHF

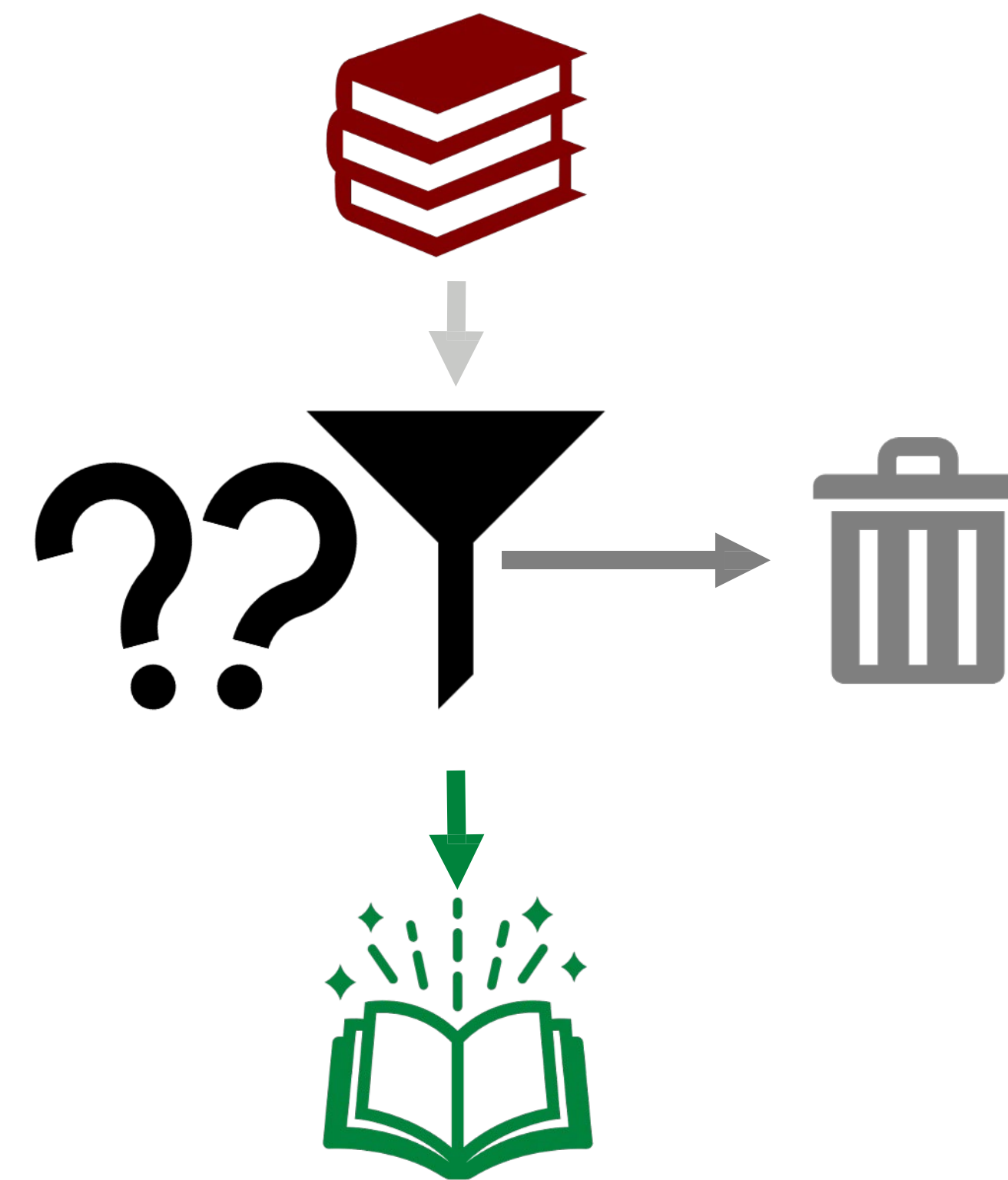
- Write demonstrations for refusing to answer
- RLHF models to prefer non-toxic generations

Safeguards at the output level

- Generate-then-classify
- Controllable text generation

Dataset filtering

- *Argument*: if you don't want your model to generate toxicity/hate speech, do not train it on such data (garbage in, garbage out)
- *Approach*: data filtering to ensure “high quality”
- How do you know what is “high quality” ?
 - GPT-2: Reddit “Karma” score as signal
 - T5, BERT: “blocklist” of “bad words”
 - GPT-3: “quality” classifier
 - Newer works: Combination of all



Blocklist of “bad” words

- “List of Dirty, Naughty, Obscene, or Otherwise Bad Words” originally by Shutterstock employees
 - Meant to prevent words in autocomplete settings
- Has been used by most companies creating LLMs
 - BERT, T5, GPT-2, etc.
- If document contains a “bad” word, remove it from training data
 - F*ck, sh*t, sex, vagina, viagra, n*gga, f*g, b*tch, etc.
- *Let’s discuss*: what are issues with this?
 - Strong risk of over-deleting bio, legal, minority content

≡ WIRED

SUBSCRIBE

TOM SIMONITE

BUSINESS FEB 4, 2021 7:00 AM

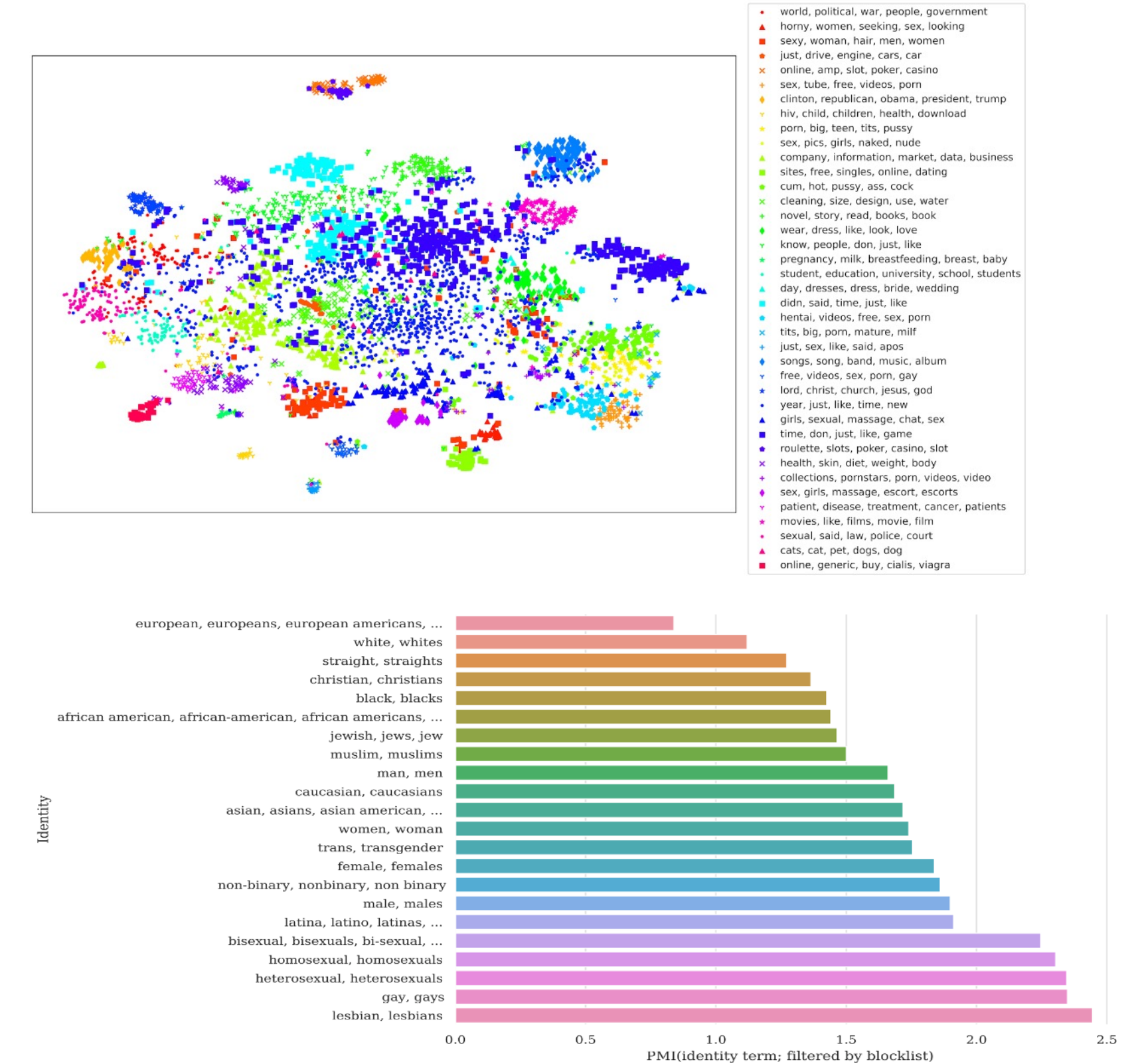
AI and the List of Dirty, Naughty, Obscene, and Otherwise Bad Words

It started as a way to restrict autocompletes on Shutterstock. Now it grooms search suggestions on Slack and influences Google's artificial intelligence research.



Effect of “bad word” blocklist filtering

- Dodge et al examined the effect of blocklist filtering on the C4 corpus
 - Found only 31% related to porn/explicit sex
 - Remaining was biology, medicine, legal
- Also examined the effect on which minority identities were removed
 - Found queer/LGBTQ identity terms removed more
- Examined dialects removed due to “bad words”
 - Found AAE, Hispanic English more likely to be removed



Less likely to be removed

- White-aligned English (6%)
- Other English (7%)

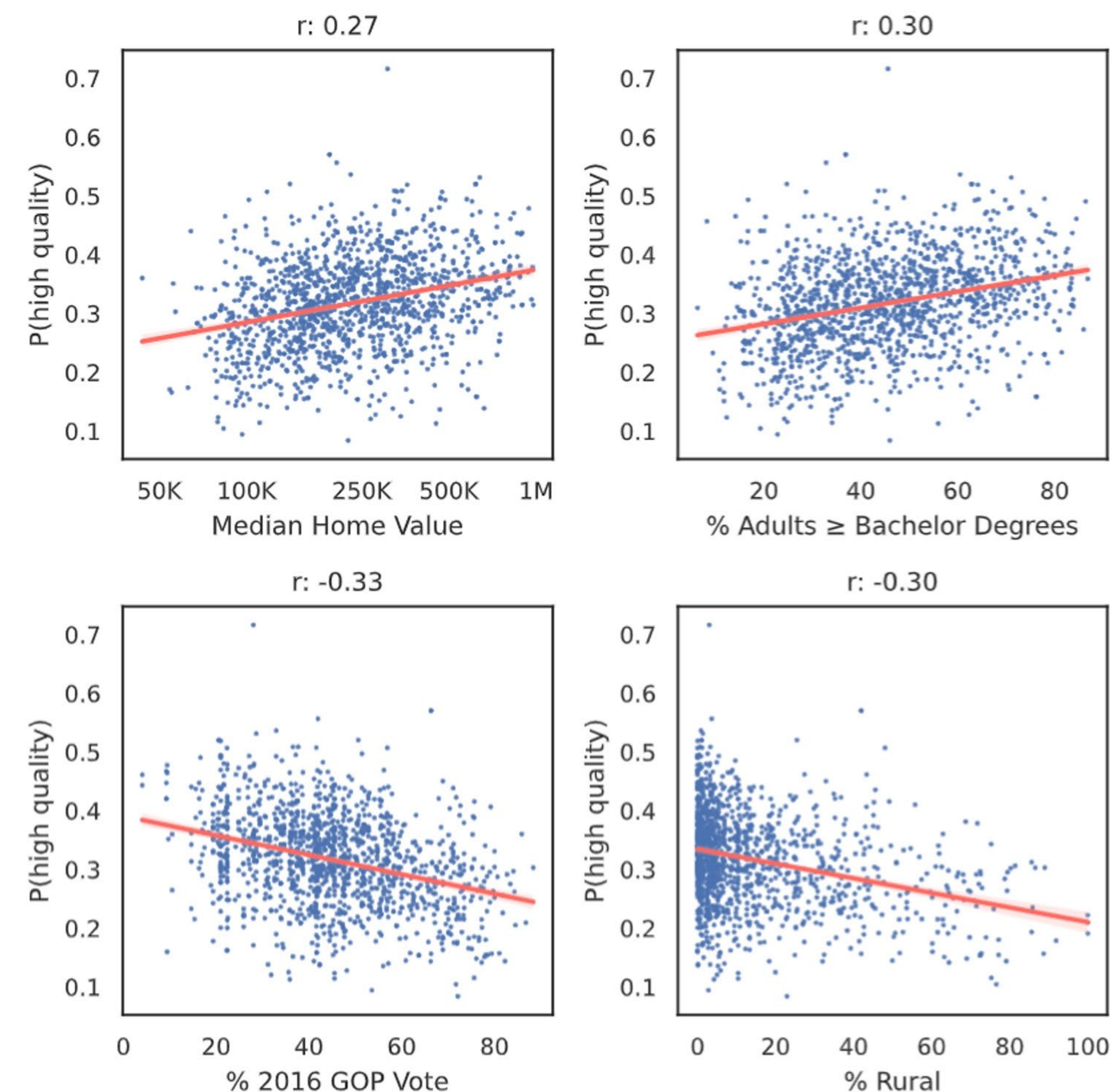
More likely to be removed

- African-American English (42%)
- Hispanic-aligned English (32%)

GPT3 Quality filter backfires

- GPT3 quality filter: similar to GPT2 data
- [Gururangan et al. \(2022\)](#) re-implemented GPT-3 quality filter
- Ran it on articles from school newspapers, which have metadata
- Filter assigns higher quality to articles from
 - Richer counties 💰
 - Counties with more educated adults 🎓
 - More liberal counties 🗳️
 - More urban counties 🏙️
- Raises language ideology question: Whose English is “good English”?

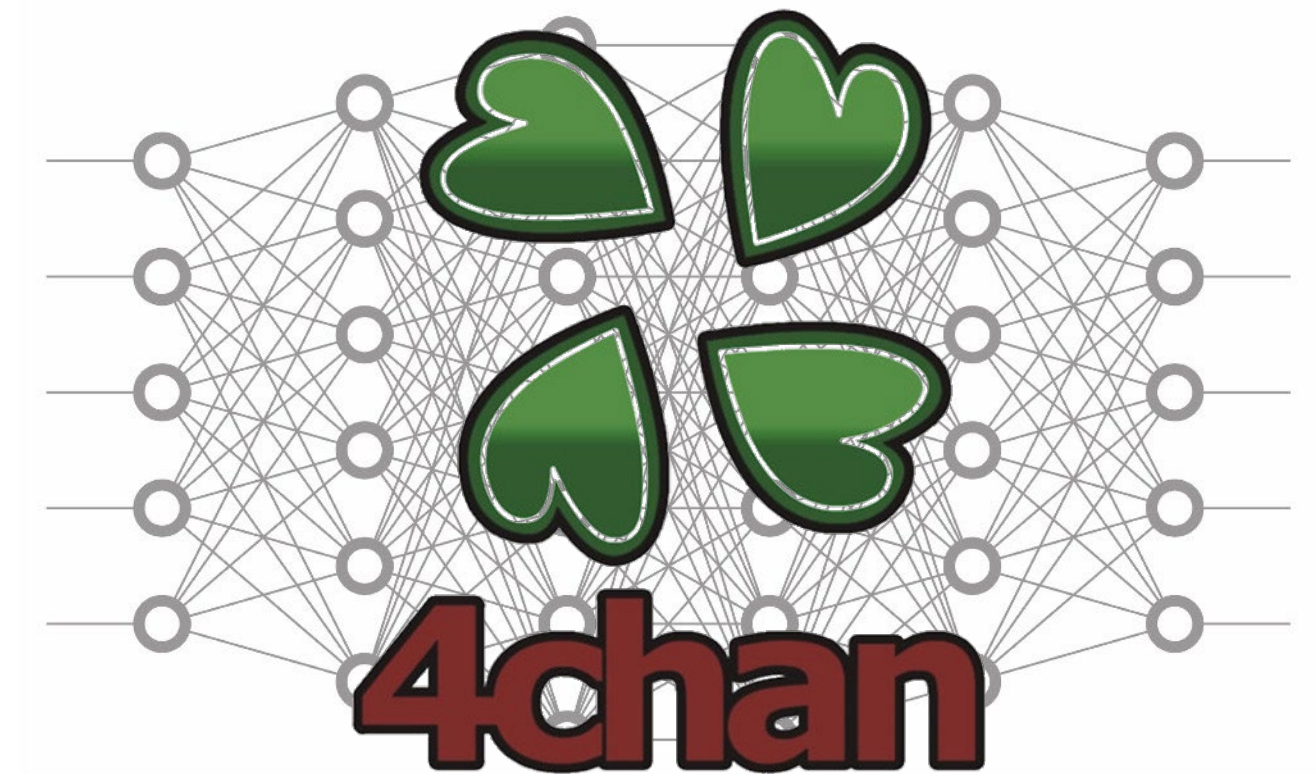
“In order to improve the quality of Common Crawl, we developed an automatic filtering method to remove low quality documents. Using the original WebText as a proxy for high-quality documents, we trained a classifier to distinguish these from raw Common Crawl.” – Brown et al. 2020



So... maybe filtering isn't a good idea since it'll backfire?

GPT4Chan controversy

- Yannic Kilchner finetuned GPT-J on 4chan posts
 - Trained on subforum /pol/ known to contain racist, sexist, white supremacist, antisemitic, anti-Muslim, anti-LGBT views
- Trolled 4chan users with bots powered by his model
 - 30,000 posts over the span of a few days
- Faced massive criticism
 - initially hosted on Huggingface, was taken down quickly
- *Let's discuss...*
 - Was this an ethical model to train? Given that the dataset was publicly available?
 - Was deploying the bots on 4chan okay?
 - Are there any useful/positive applications of the model?



≡ FORTUNE

TECH · 4CHAN

‘This breaches every principle of human research ethics’: A YouTuber trained an A.I. bot on toxic 4Chan posts then let it loose — and experts aren’t happy

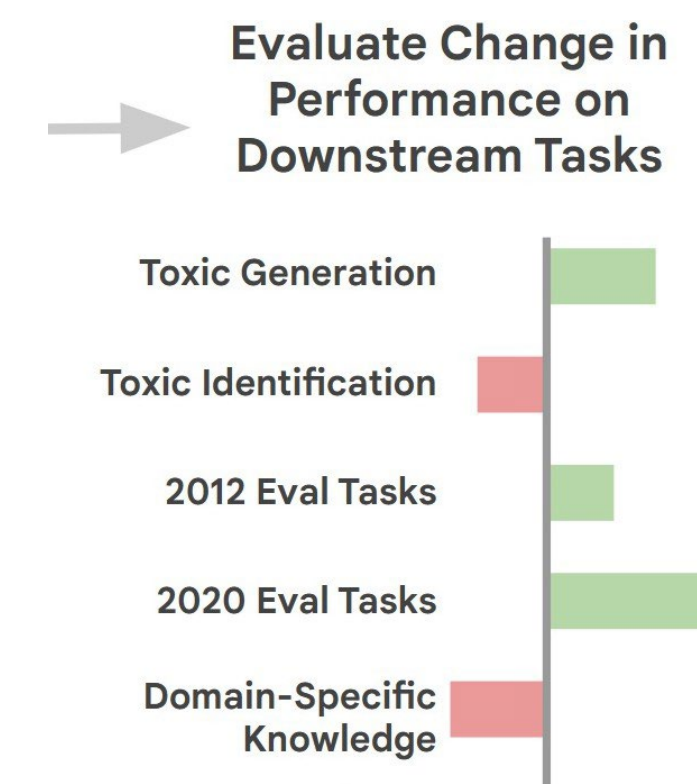
BY SOPHIE MELLOR

June 10, 2022 at 5:23 AM EDT

<https://thegradient.pub/gpt-4chan-lessons>

Why LLMs might want to have seen toxic content

- Detecting hate speech [[Chiu et al 2022](#)]
 - [Longpre et al. \(2023\)](#) showed that LLMs trained on more toxicity are better toxicity detections
 - Improving hate speech models with data augmentation: ToxiGen [[Hartvigsen et al 2022](#)]
- Counter speech generation [[Saha et al 2022](#), [Kim et al 2022](#), [Mun et al 2023](#)]
- *If we train on toxicity, something else must be done at a different time!*



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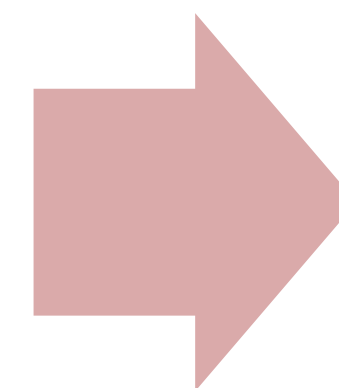
Safeguards at the output level

- Generate-then-classify
- Controllable text generation

RLHF safeguarding – assumptions

- PPO & family:

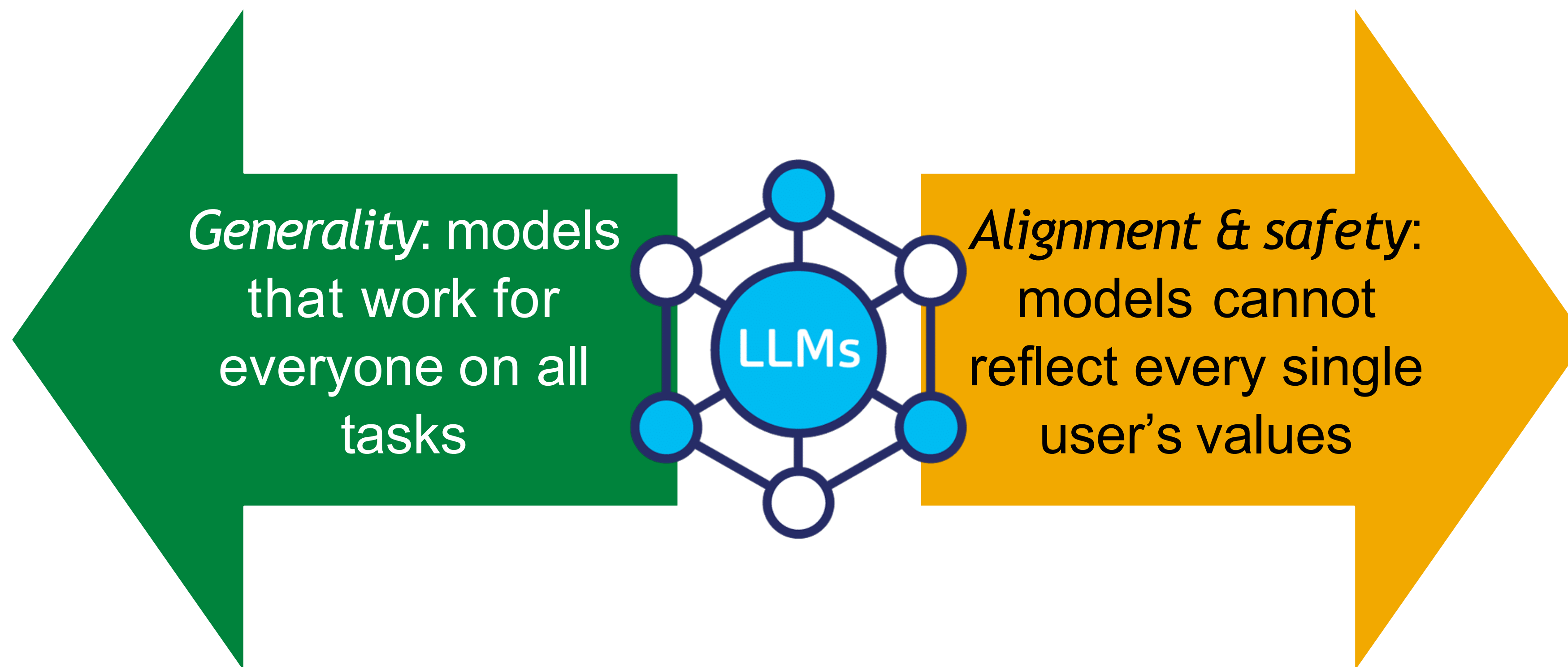
Obtain preference data:
which generation is good vs.
bad?



RL is done to encourage
more like “preferred output”

- Big question: what does it mean for a generation to be better/preferred?
 - How to balance harmless and helpful? [[Bai et al '23](#)]
 - *E.g., “help me create a poisonous drink.”*
 - What if people’s preferences are biased or gameable?
 - *E.g., people prefer certainty over uncertainty in answers to questions [[Zhou et al. 24](#)]*
 - Fundamental issue: cannot represent all values and cultures into one ranking.
 - *Casper et al. 2023. “Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback.” arXiv [cs.AI]. arXiv. <http://arxiv.org/abs/2307.15217>*

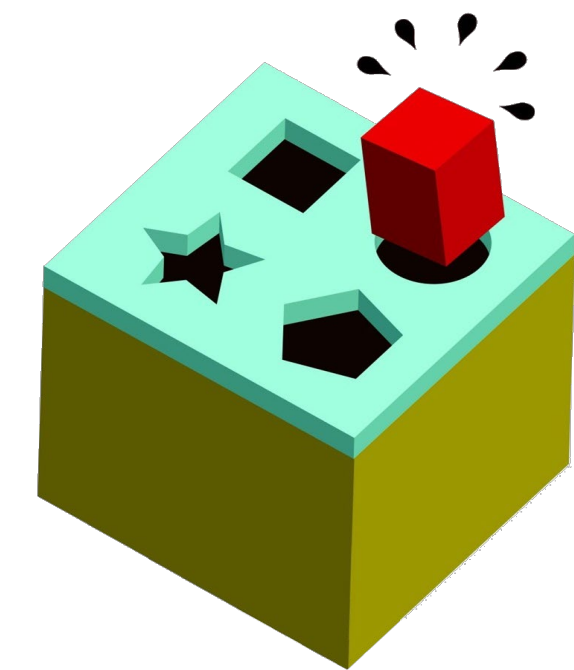
Big unresolved tension



- Let's discuss: what do y'all think we should do?
 - It's complicated!

So... what can we do?

- Need to keep studying what models can and can't do, who they work for and don't work for
- Narrow scope of model users
 - Community-specific models (e.g., Masakhane Initiative)
- Specialize models' abilities / away from one-size-fits-all
 - E.g., toxicity explanation generation model needs to generate stereotypes, but story generation models might not
- In line with many legislative efforts: legislate the application or task, not the model



Takeaways

- AI systems are biased
 - Real world is biased, data is biased
 - ML objectives play a role
 - Annotation interfaces, context plays a role
 - Debiasing is challenging, requires socio-technical lens
- Toxicity and undesirable content
 - Longer-tail phenomenon, present in training data
 - Filtering data can backfire
 - Safeguarding to all people is impossible
- *Any questions?*

